

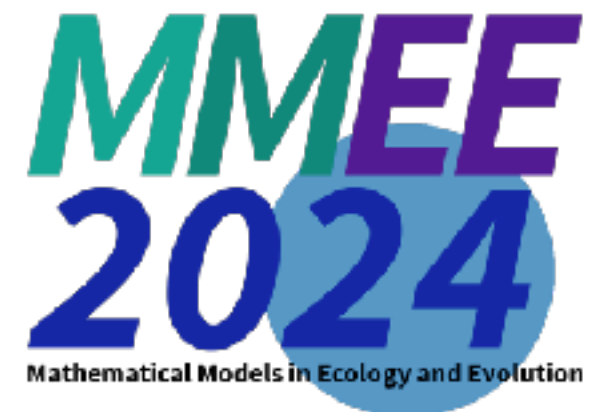


The role of phenotypic plasticity in the ecological theatre

Ata Kalirad

Department for Integrative Evolutionary Biology

July 17th, 2024



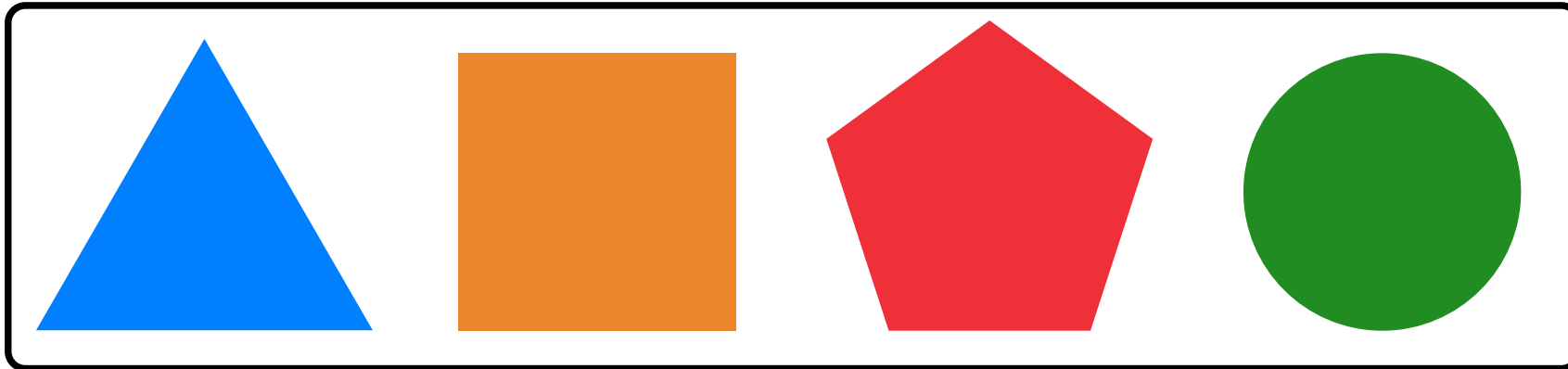
The puzzle of community assembly



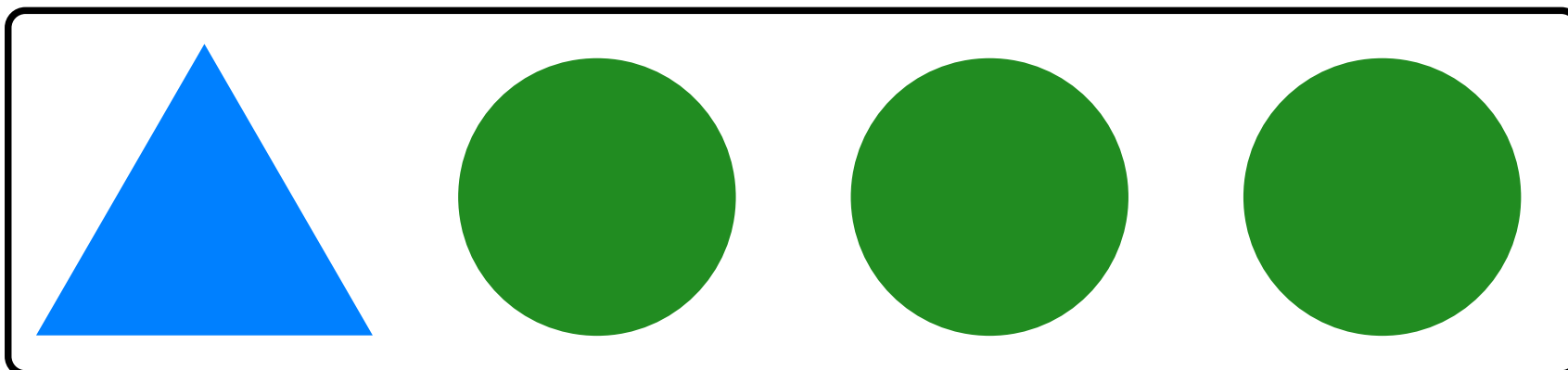
Reserva da Biosfera da Mata Atlântica (Source: unesco.org)

The puzzle of community assembly

Initial community composition

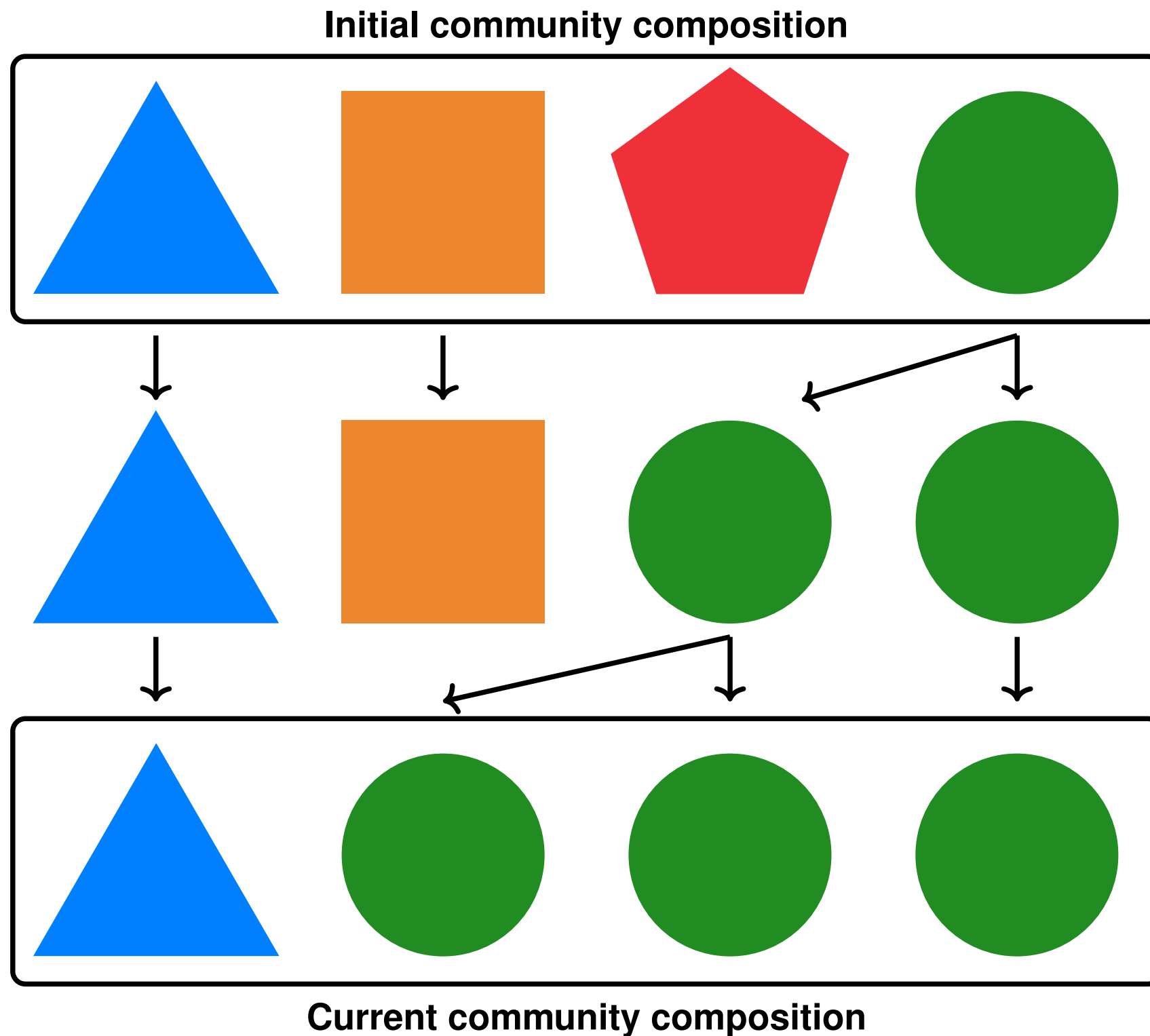


Past



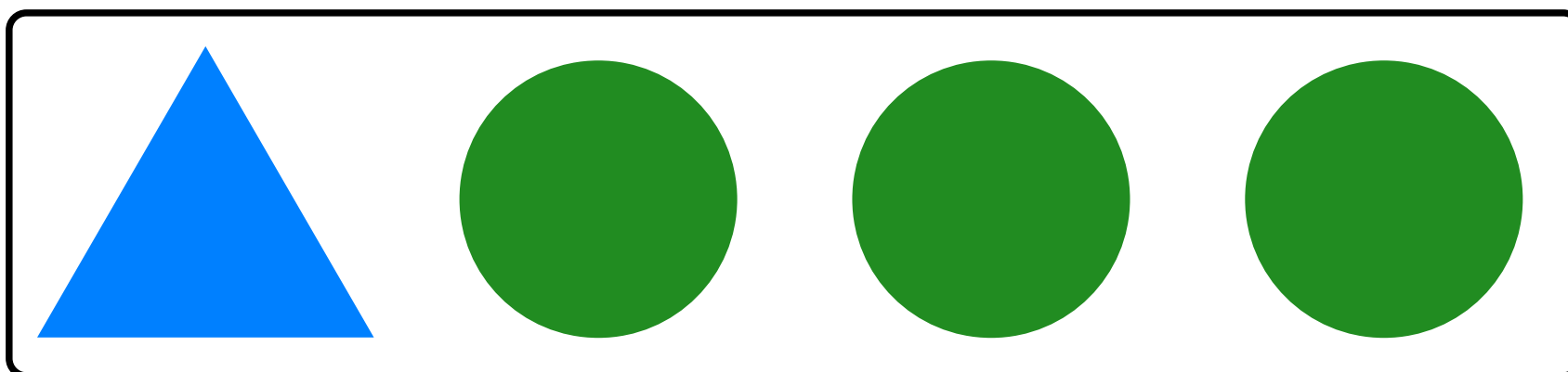
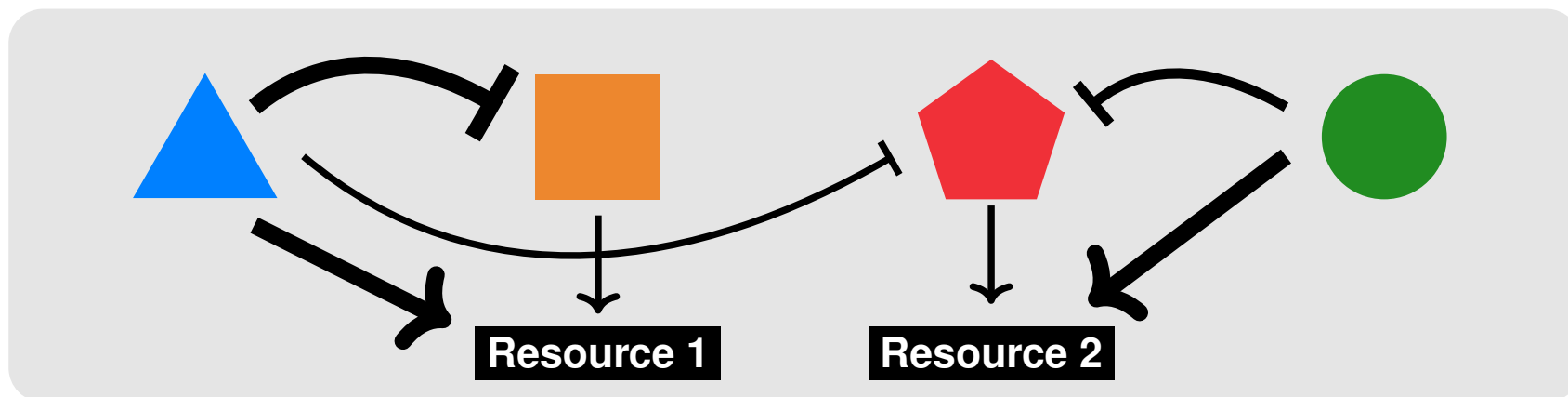
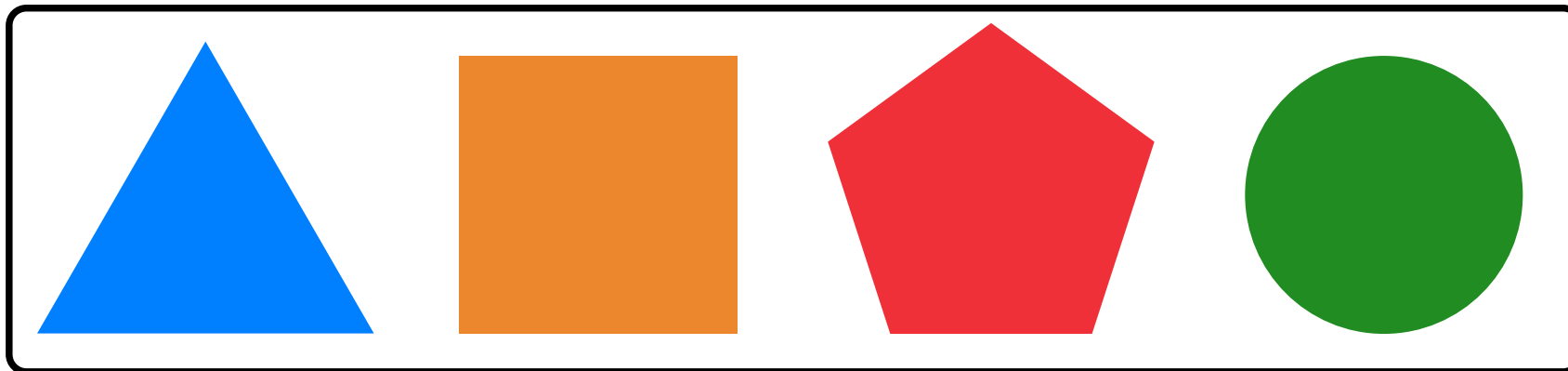
Current community composition

The puzzle of community assembly



The puzzle of community assembly

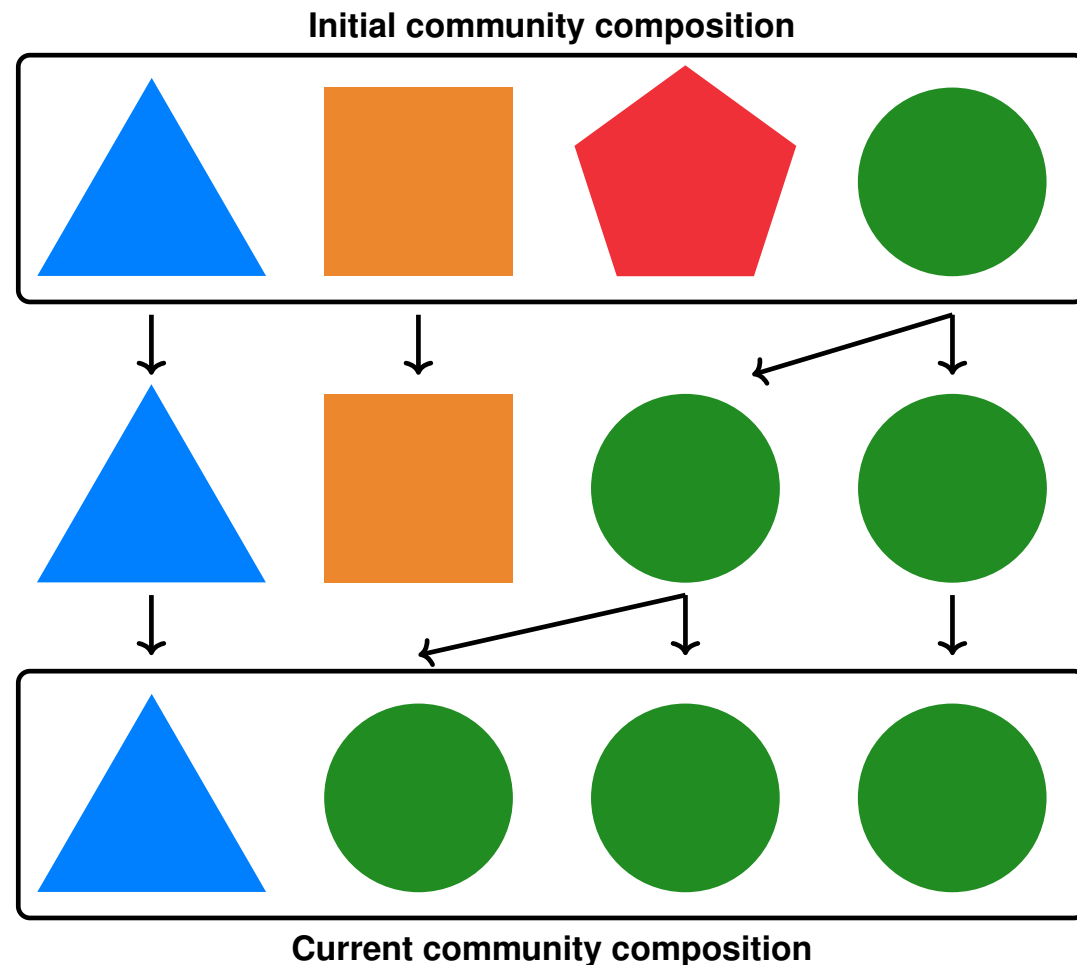
Initial community composition



Current community composition

The puzzle of community assembly

Stochasticity



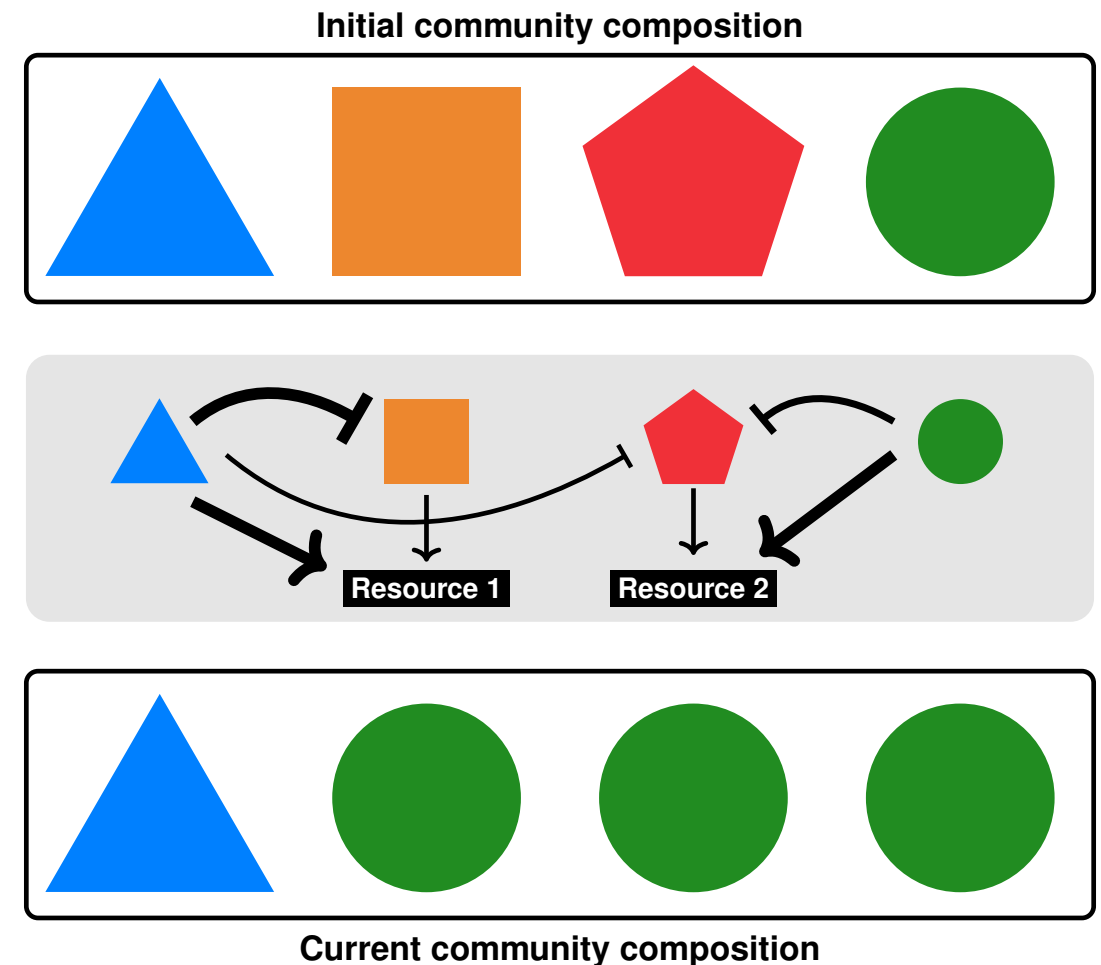
Riley, G. A. 1963. Marine biology: Proceedings of the first international interdisciplinary conference. American Institute of Biological Sciences, Washington, DC.

MacArthur, R. H., & Wilson, E. O. (1963). An Equilibrium Theory of Insular Zoogeography. *Evolution*, 17(4), 373–387.

Simberloff, D. (1980). A Succession of Paradigms in Ecology: Essentialism to Materialism and Probabilism. *Synthese*, 43(1), 3–39.

Hubbell, S. P. 2001. The unified neutral theory of biodiversity and biogeography. Princeton University Press, Princeton, NJ.

Determinism



MacArthur, R., & Levins, R. (1964). COMPETITION, HABITAT SELECTION, AND CHARACTER DISPLACEMENT IN A PATCHY ENVIRONMENT. *PNAS*, 51(6), 1207–1210.

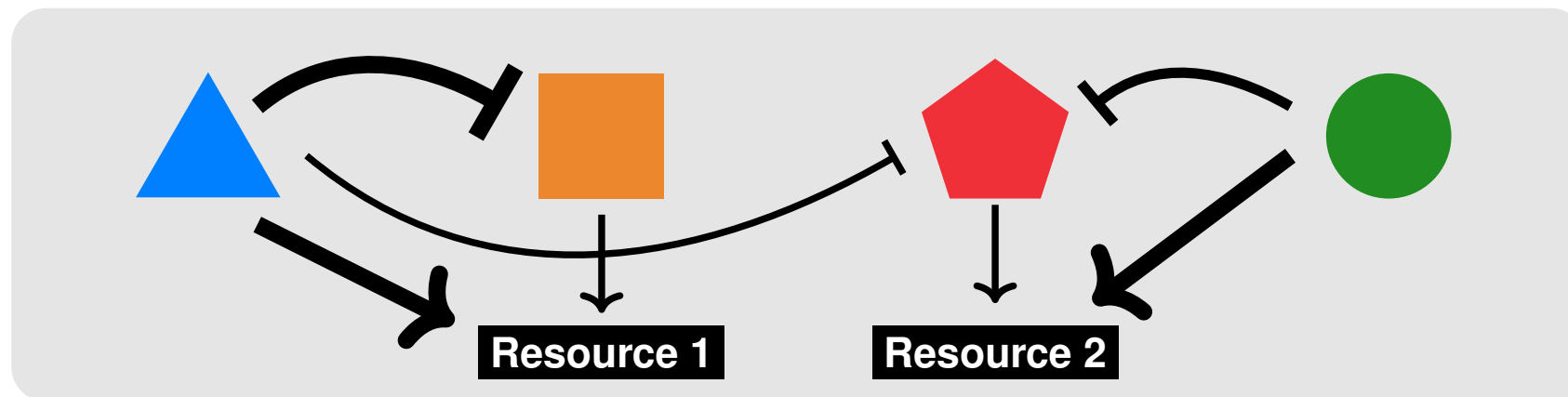
MacArthur, R., and R. Levins. 1967. The limiting similarity, convergence, and divergence of coexisting species. *American Naturalist* 110:377–385.

Cody ML, Diamond JM, ed. (1975). Ecology and Evolution of Communities. Cambridge, MA: Belknap Press

Chesson, P. L. 2000. Mechanisms of maintenance of species diversity. *Annual Review of Ecology and Systematics* 31:343–366.

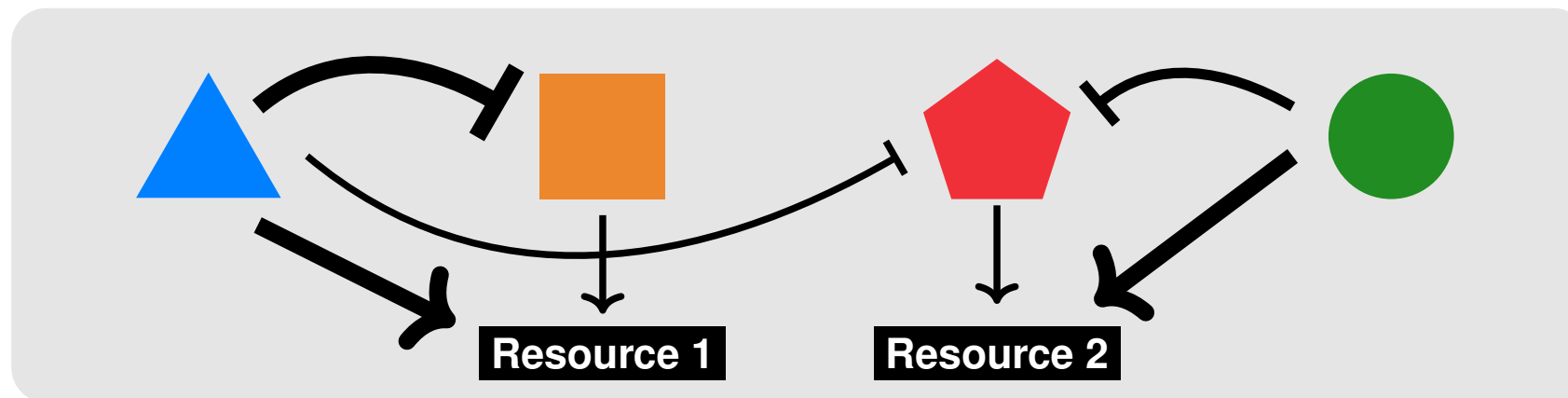
A new wrinkle

The sinuous road between genotype and phenotype

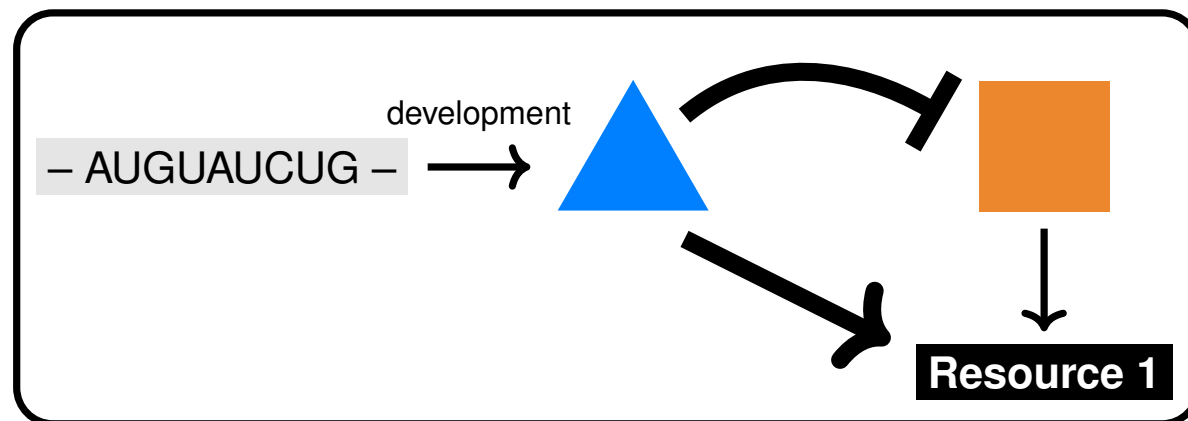


A new wrinkle

The sinuous road between genotype and phenotype

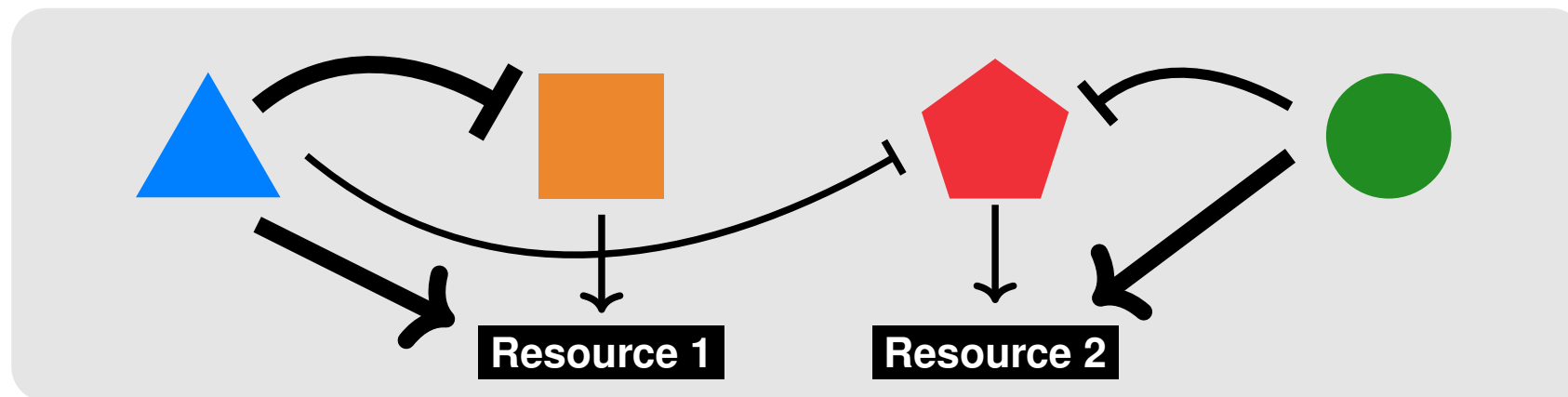


Environment A

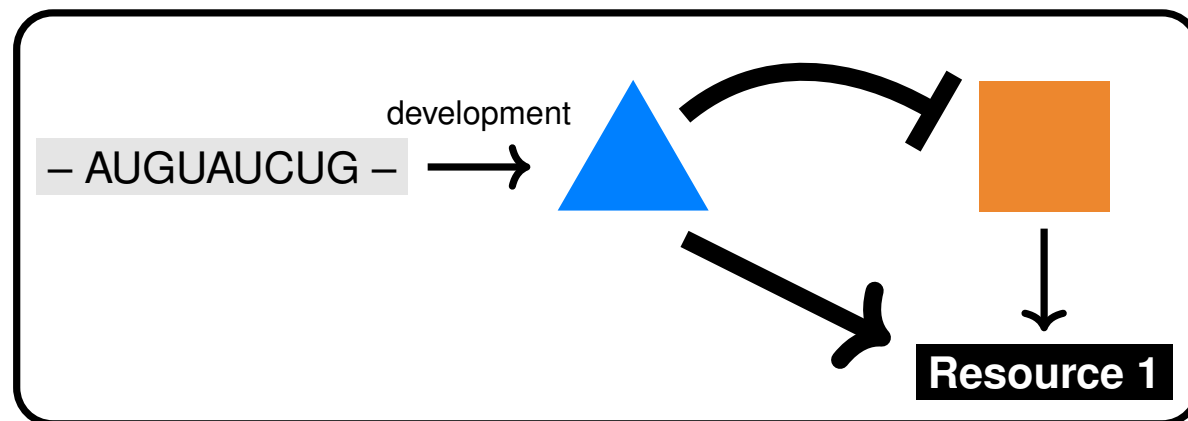


A new wrinkle

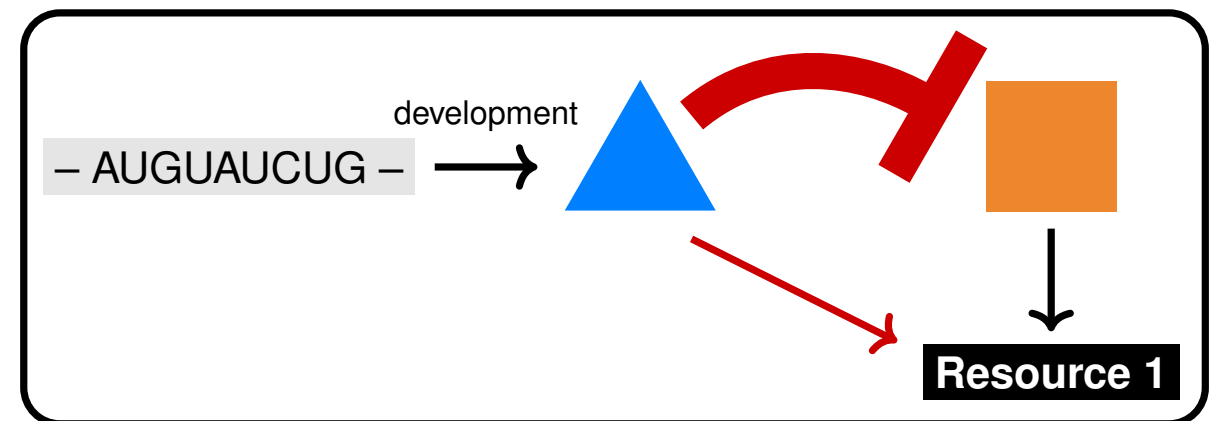
The sinuous road between genotype and phenotype



Environment A

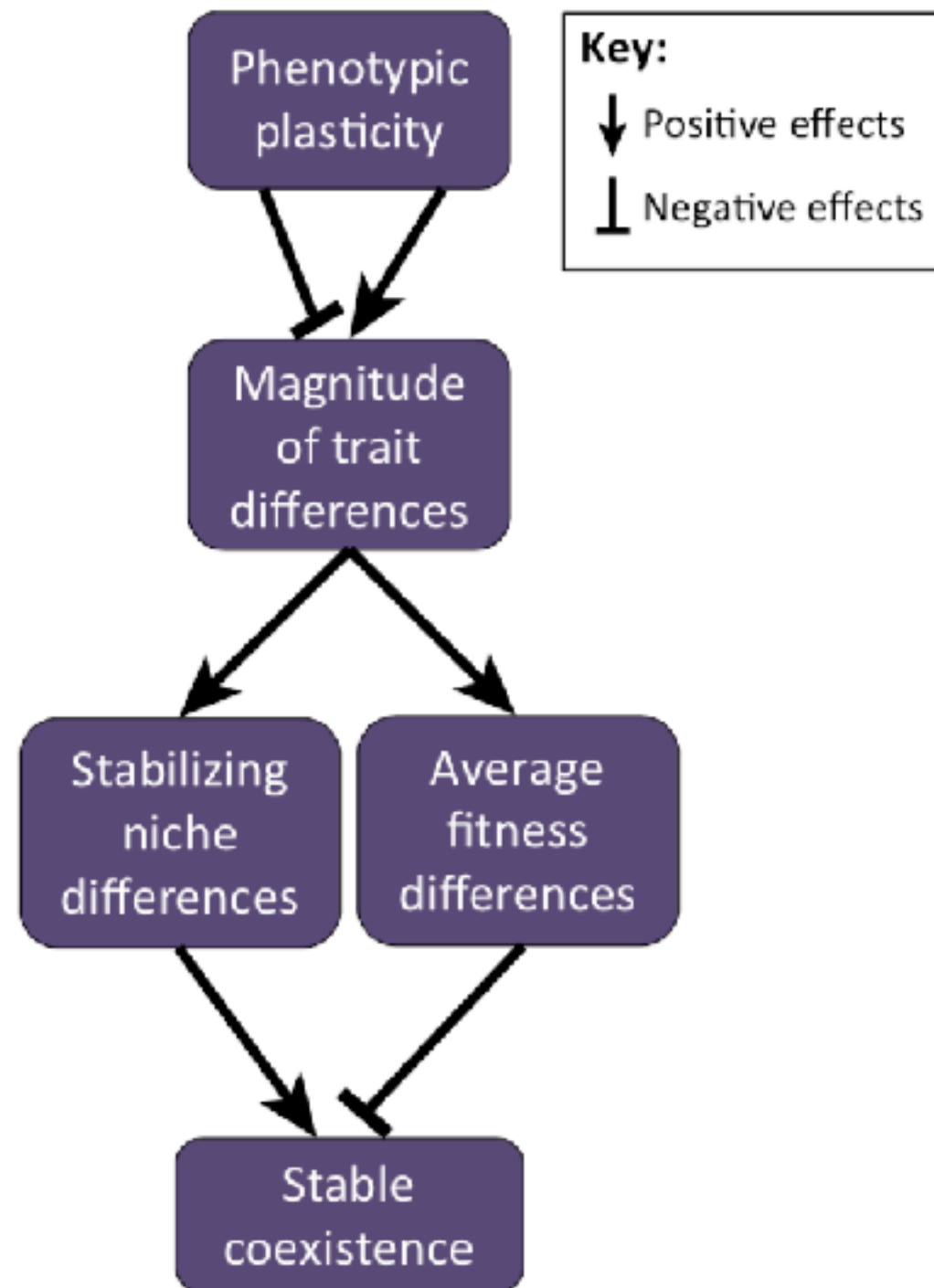


Environment B



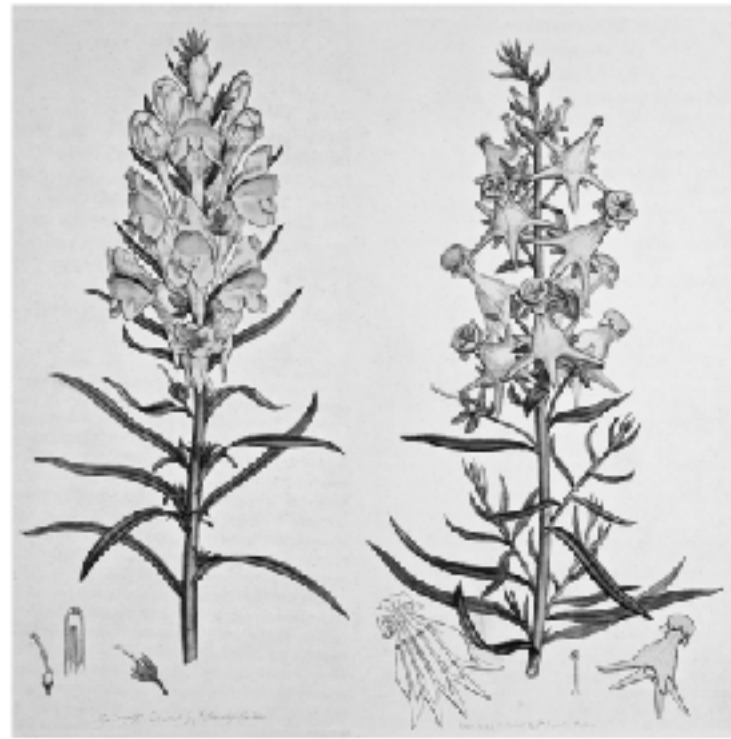
A new wrinkle

Phenotypic plasticity and community assembly



A new wrinkle

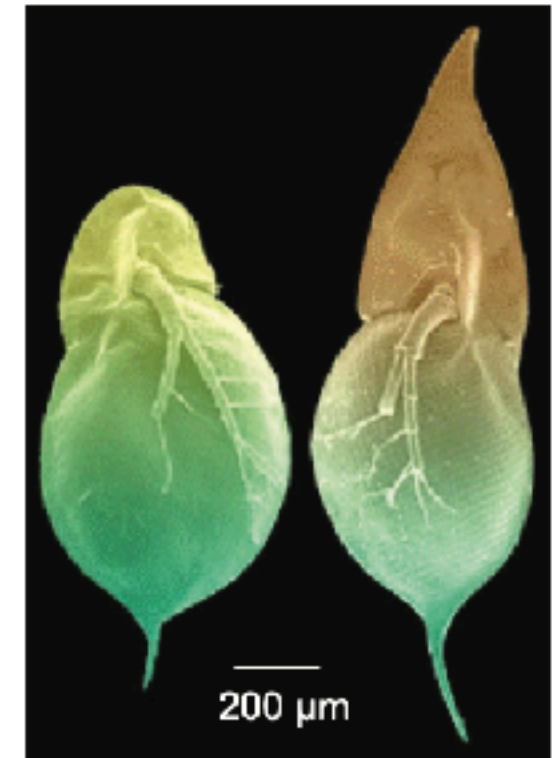
Phenotypic plasticity across scales



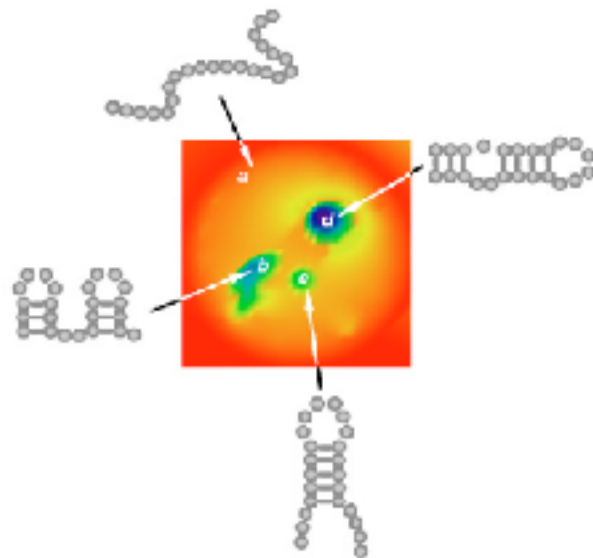
Linaria vulgaris (from Bounduriasky and Day 2018)



Nijhout 2002



Pfennig 2021

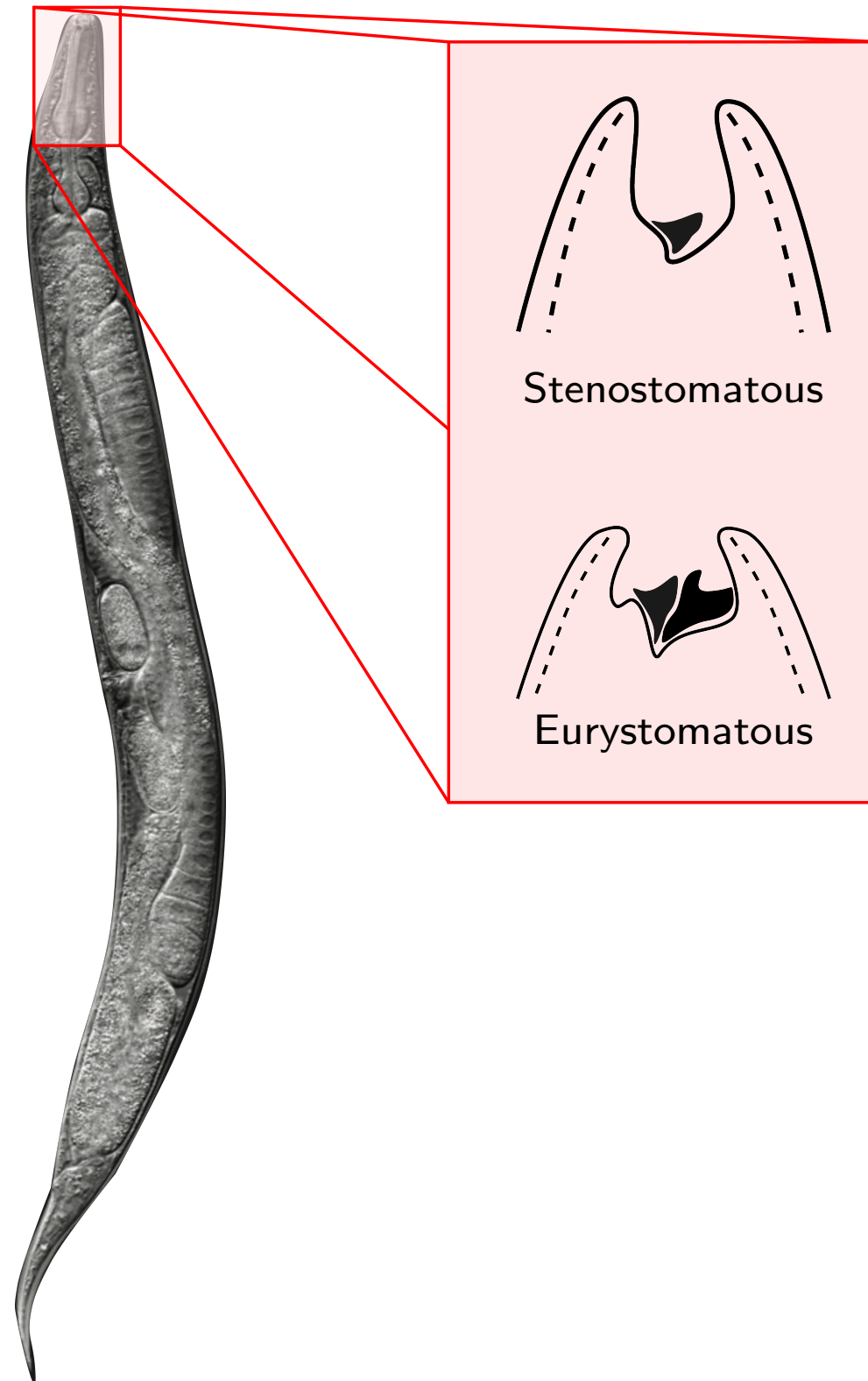


Godet et al. 2012

Pristionchus pacificus



A perfect model of plasticity



A perfect model of plasticity

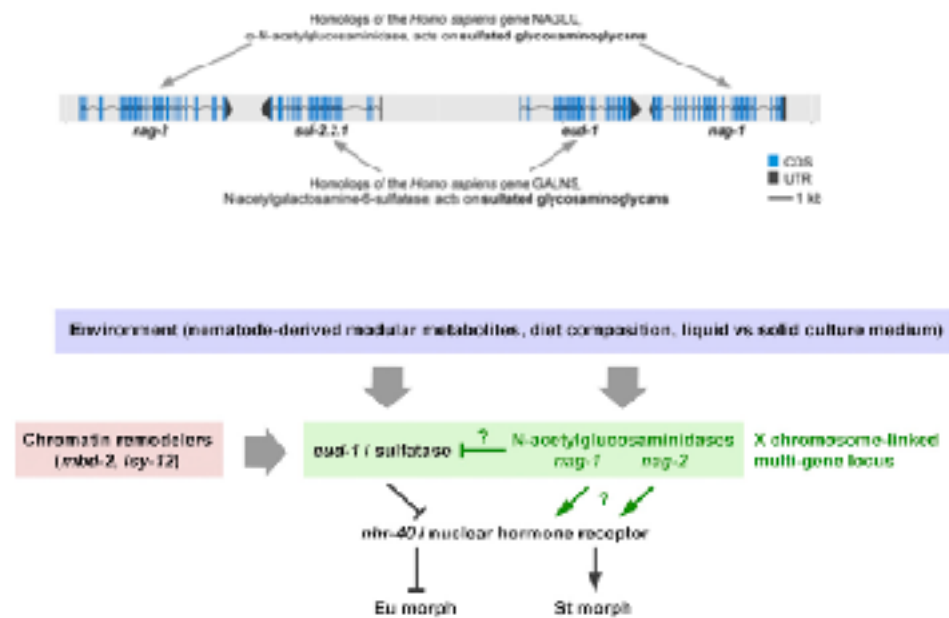
Plasticity with consequence



P. pacificus (Eu) predator killing a *C. elegans* larva

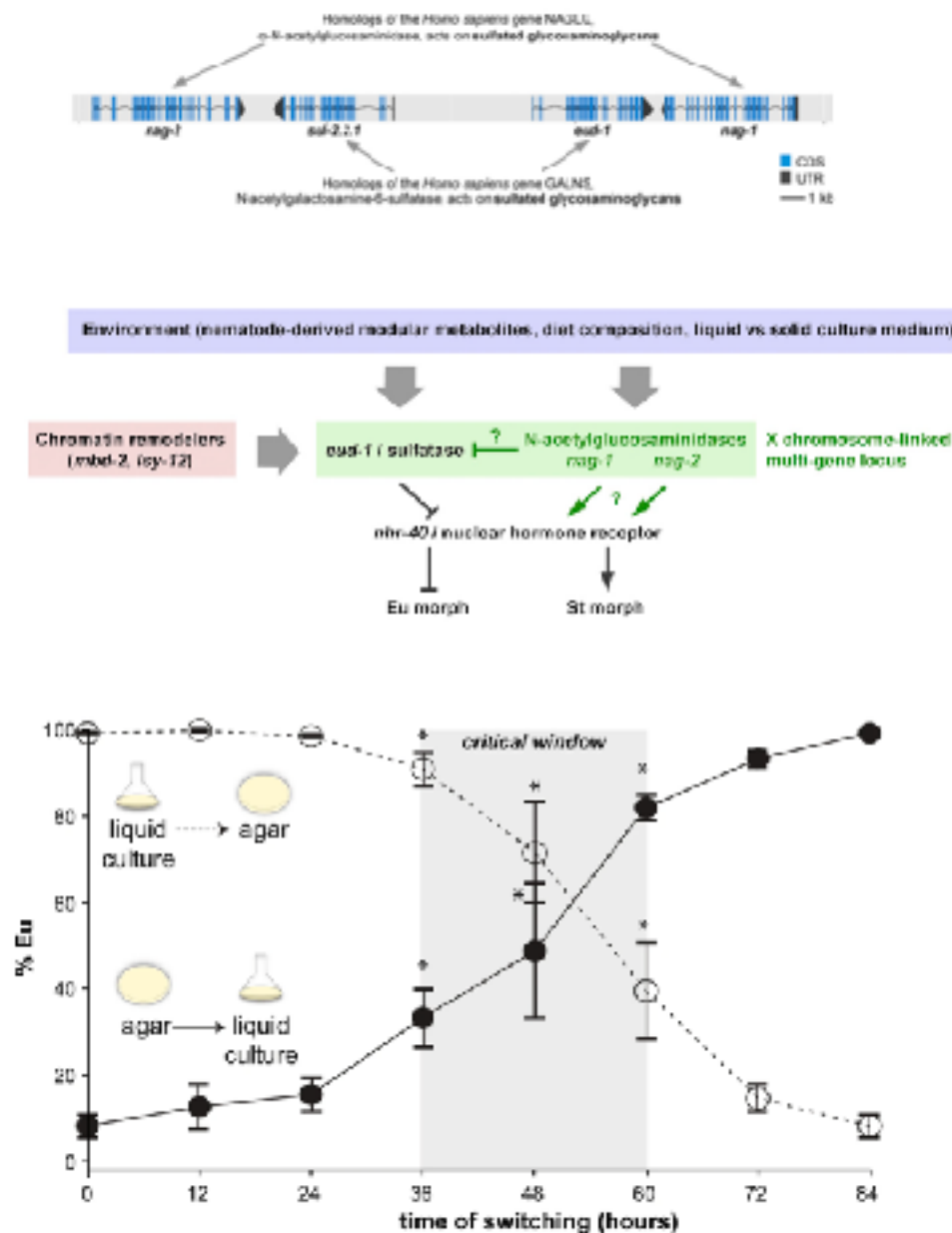
A perfect model of plasticity

Comprehensive understanding



A perfect model of plasticity

Comprehensive understanding

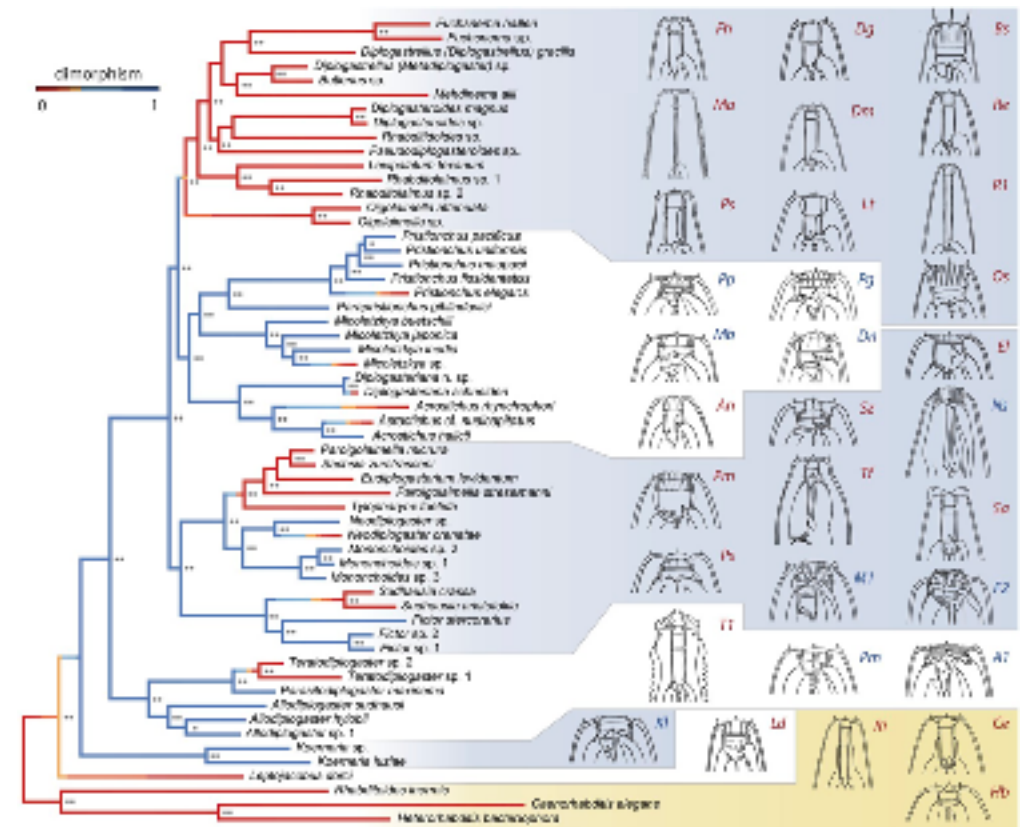
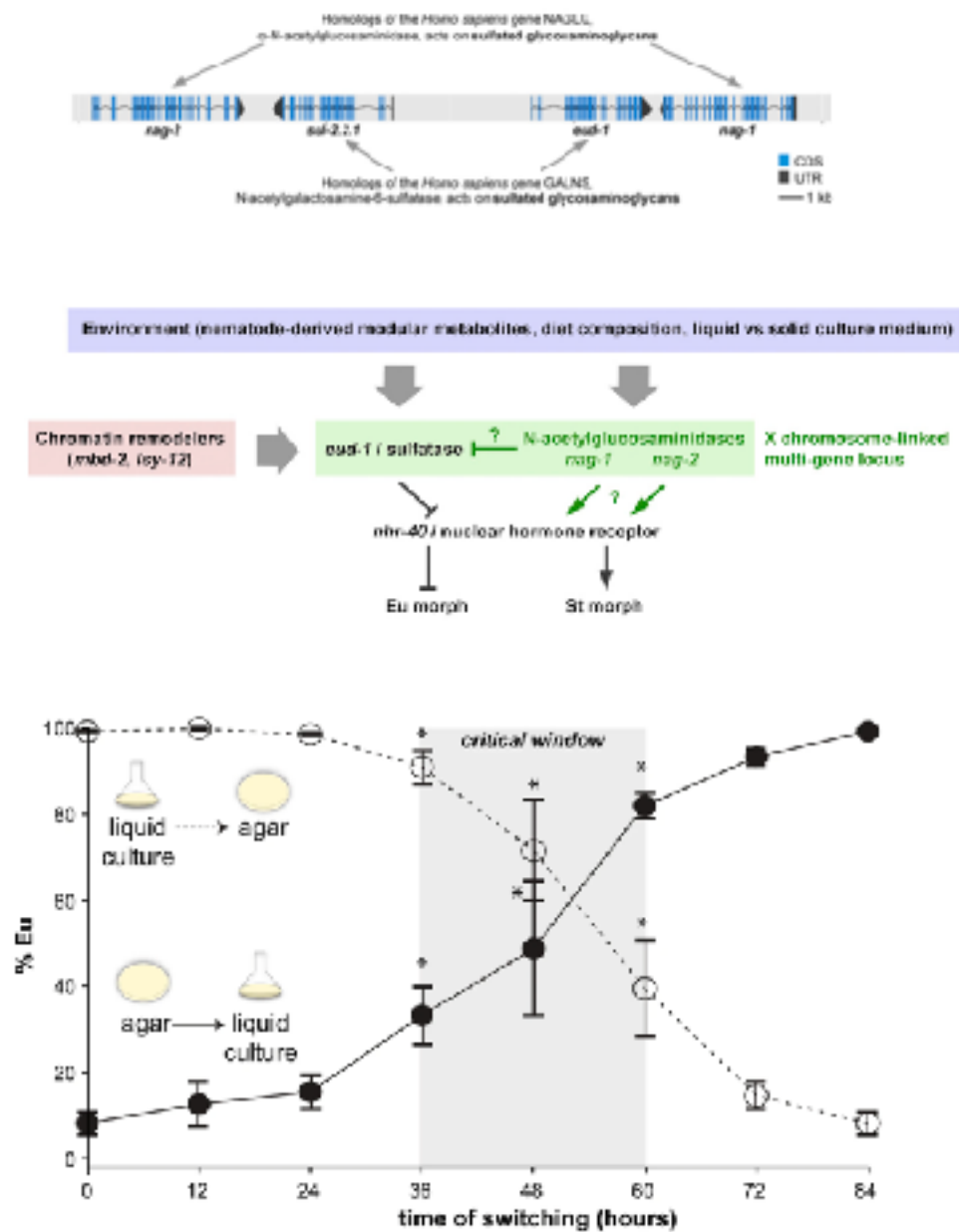


Sieriebriennikov et al. A Developmental Switch Generating Phenotypic Plasticity Is Part of a Conserved Multi-gene Locus. *Cell Rep.* 2018 Jun 5;23(10):2835-2843.e4.

Werner et al. Histone 4 lysine 5/12 acetylation enables developmental plasticity of *Pristionchus* mouth form. *Nat Commun* 14, 2095 (2023).

A perfect model of plasticity

Comprehensive understanding



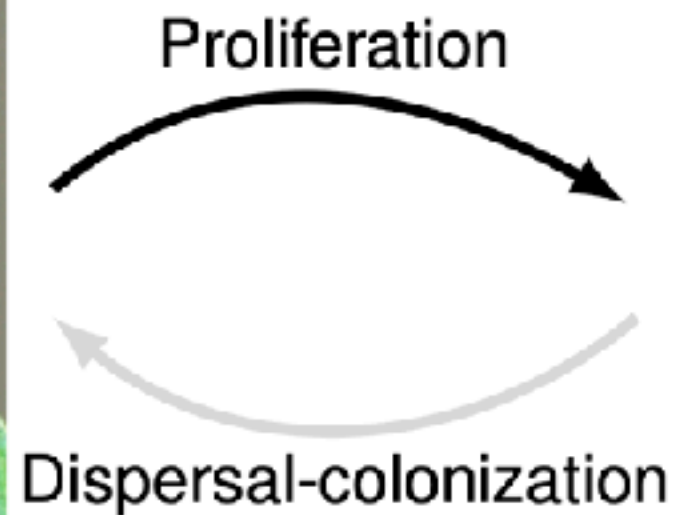
Sieriebriennikov et al. A Developmental Switch Generating Phenotypic Plasticity Is Part of a Conserved Multi-gene Locus. *Cell Rep.* 2018 Jun 5;23(10):2835-2843.e4.

Werner et al. Histone 4 lysine 5/12 acetylation enables developmental plasticity of *Pristionchus* mouth form. *Nat Commun* 14, 2095 (2023).

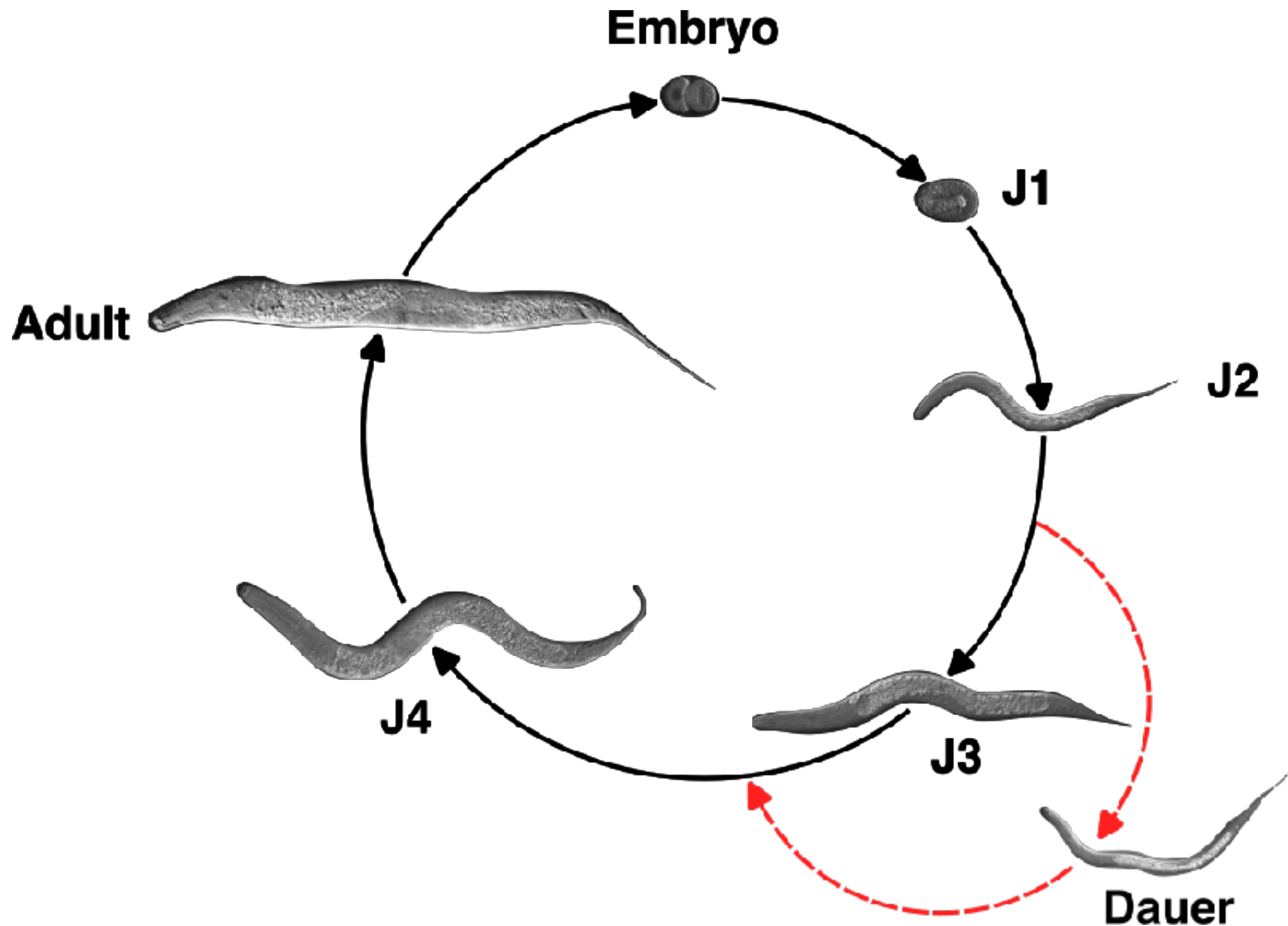
Susoy et al. (2015) Rapid diversification associated with a macroevolutionary pulse of developmental plasticity *eLife* 4:e05463.

A perfect model

Boom and bust in nature

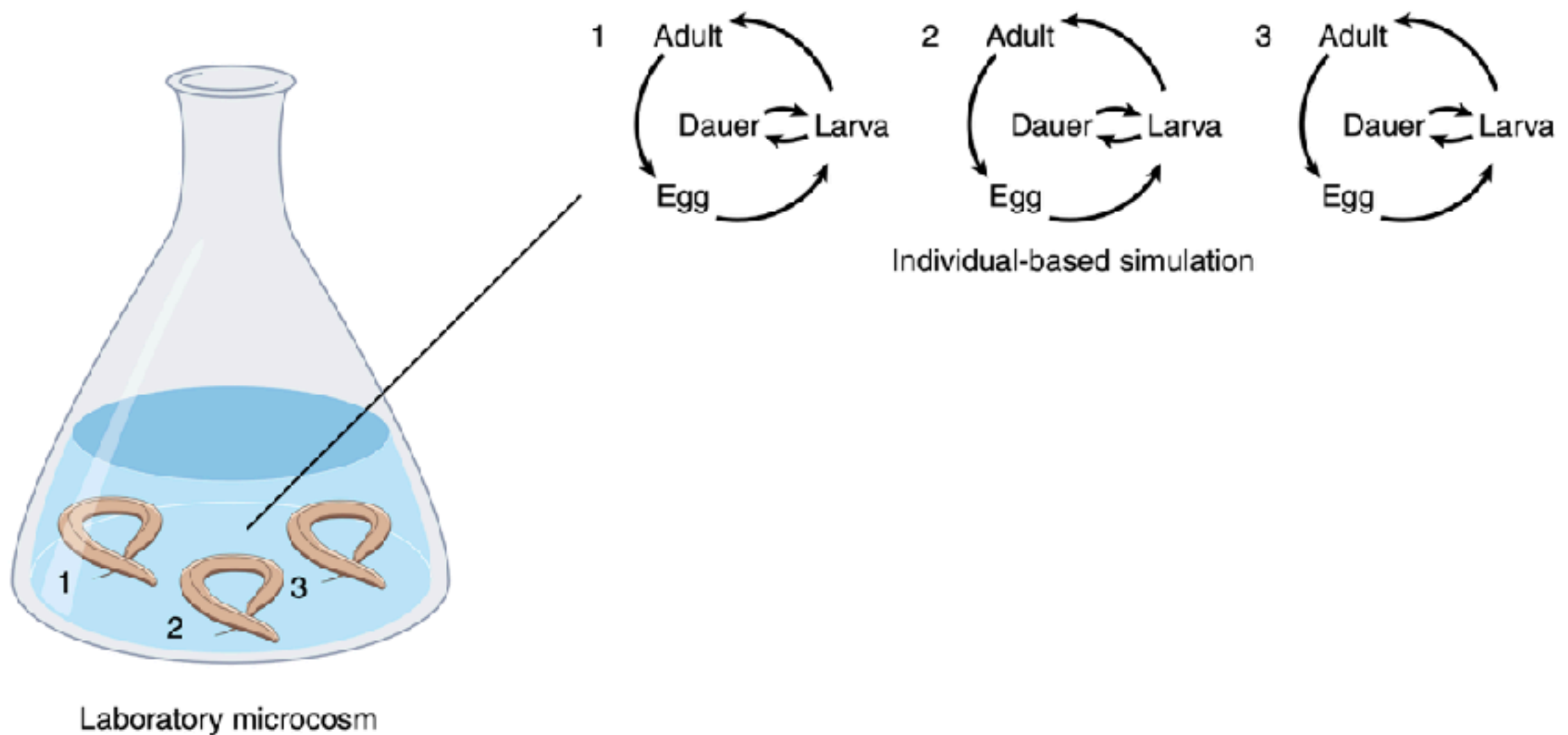


A perfect model



PriPOP model

Inspirations: IBM in *C. elegans*



Kalirad, A., Sommer, R.J. The art of mechanistic modeling in biology. *Nat Comput Sci* **2**, 72–73 (2022).

Scharf, A. et al. *Nat. Comput. Sci.* <https://doi.org/10.1038/s43588-022-00190-8> (2022).

PriPOP model

Inspirations: missing variation

Trends in Ecology & Evolution



Volume 26, Issue 4, April 2011, Pages 183-192

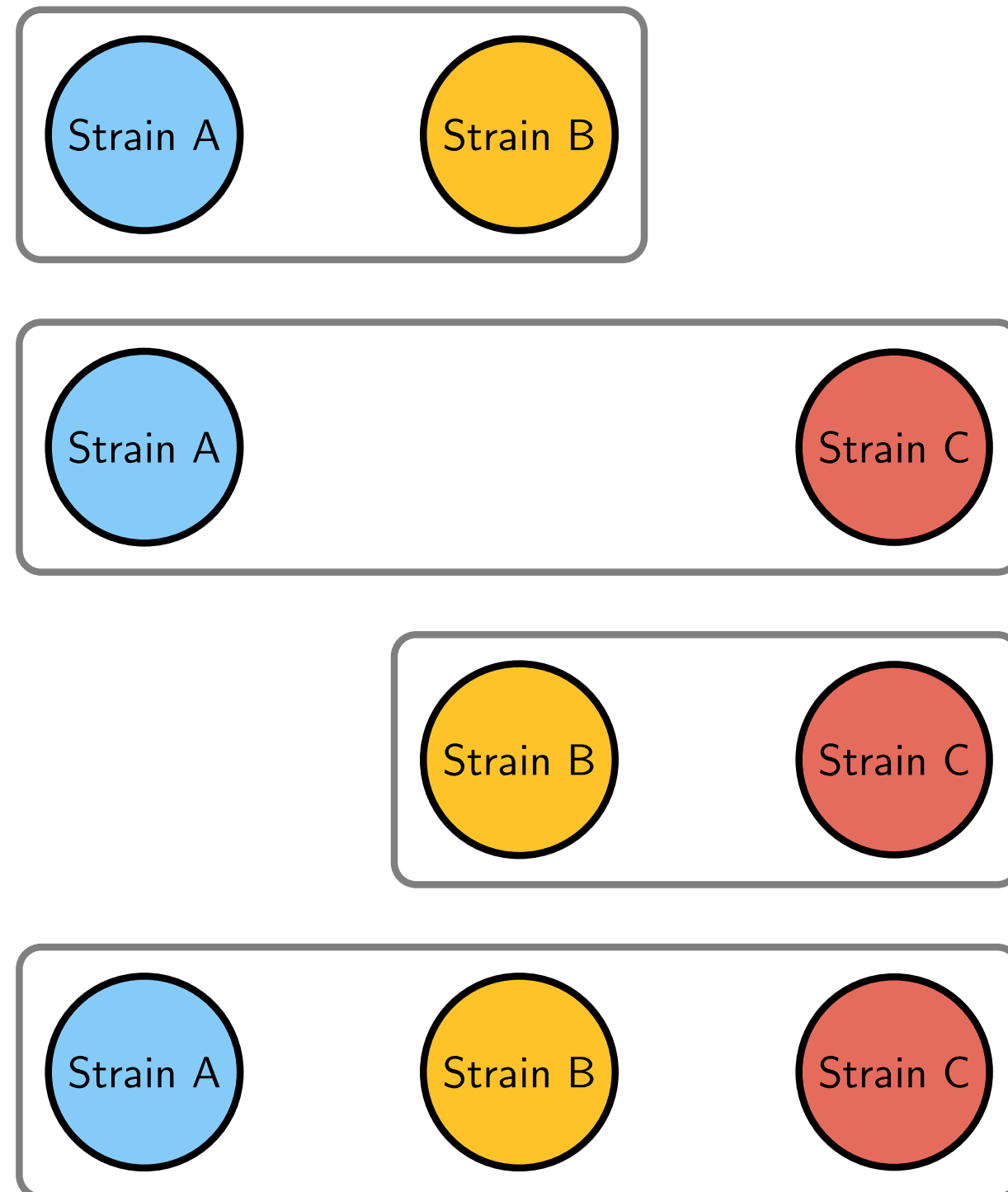
Review

Why intraspecific trait variation matters in community ecology

Daniel I. Bolnick¹ , Priyanga Amarasekare², Márcio S. Araújo³, Reinhard Bürger⁴,
Jonathan M. Levine⁵, Mark Novak⁶, Volker H.W. Rudolf⁷, Sebastian J. Schreiber⁸, Mark C. Urban⁹
, David A. Vasseur¹⁰

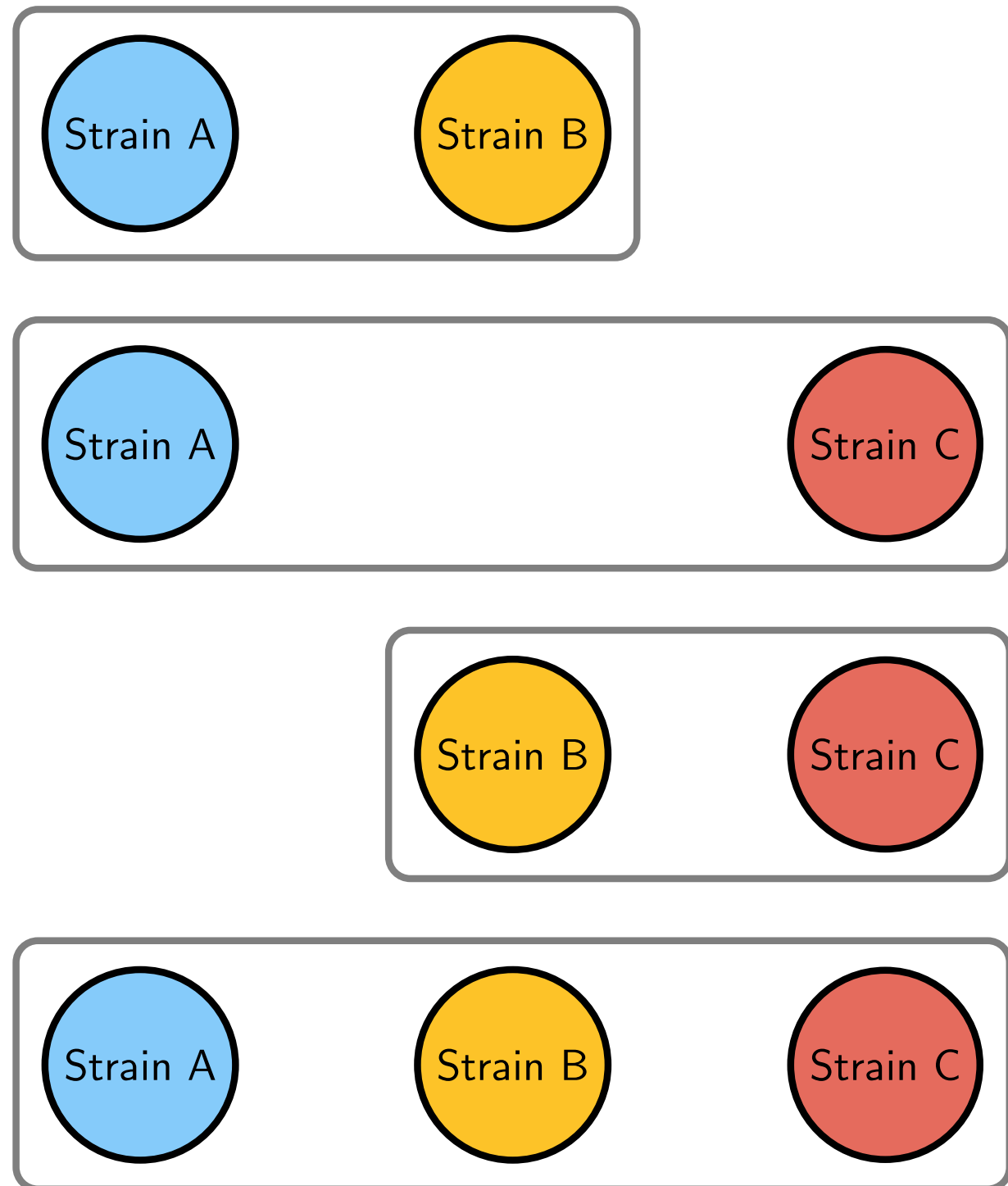
PriPOP model

Minimal communities

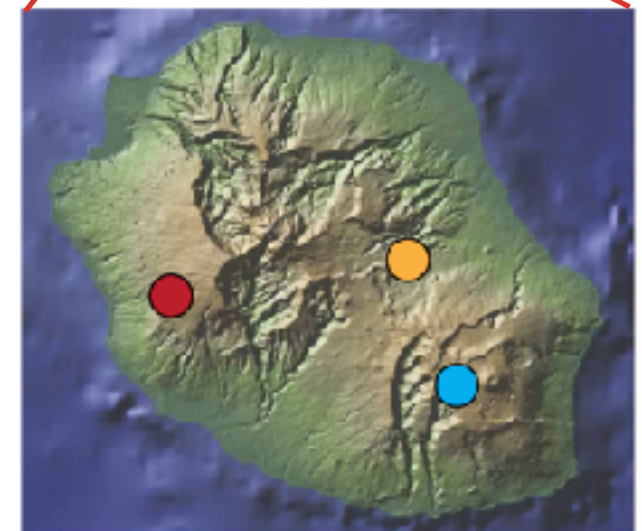


PriPOP model

Minimal communities



La Réunion



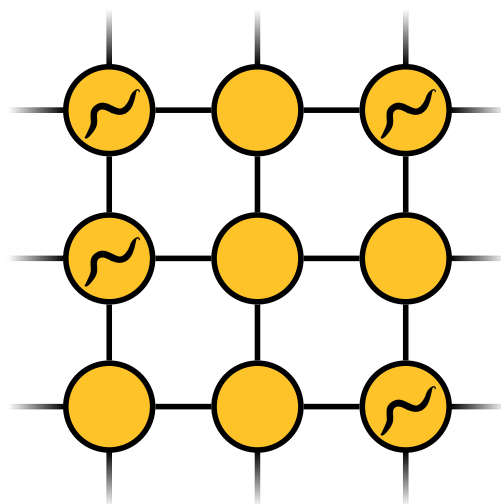
● Trois Bassins (TB) ● Grand Etang (GE) ● Nou du Bassin (NB)

PriPOP model

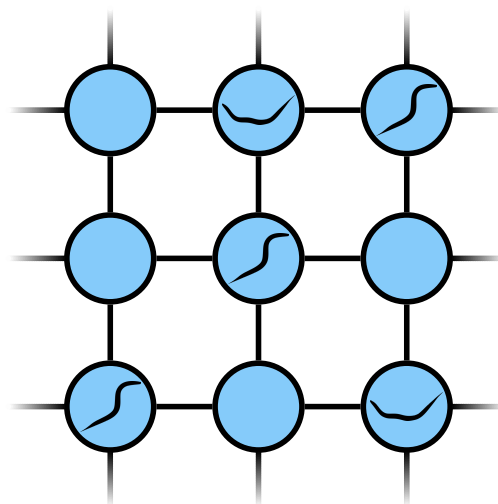
Spatially-explicit IBM

a

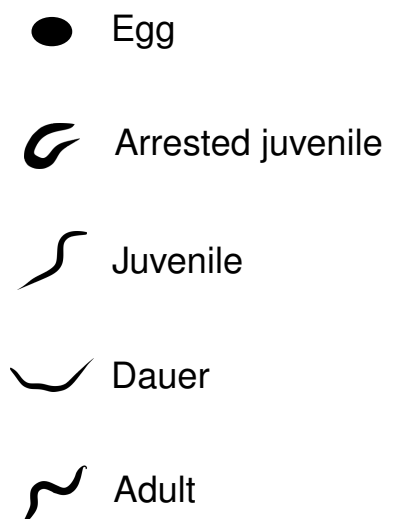
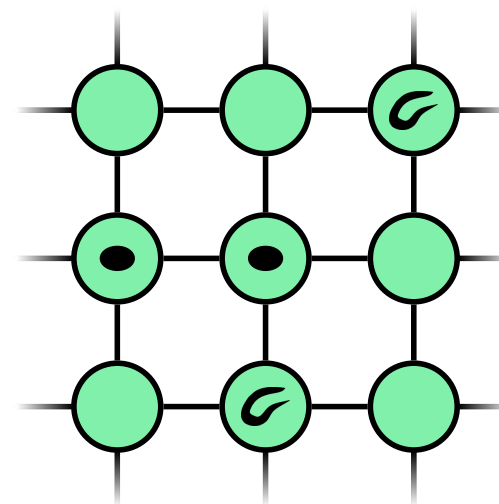
Top layer



Middle layer

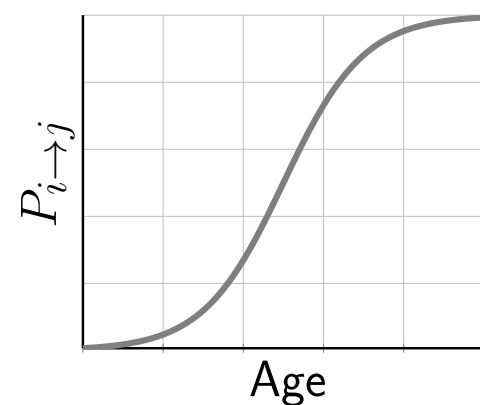


Bottom layer

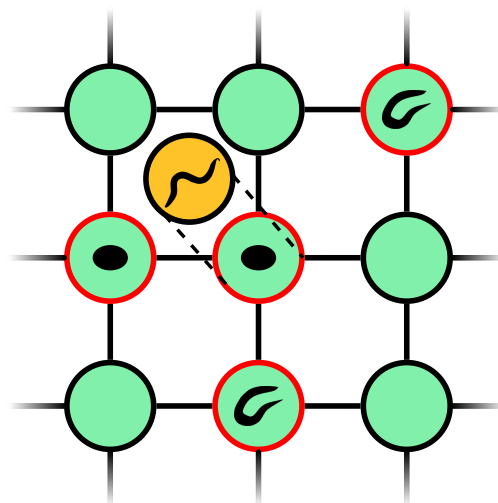


b

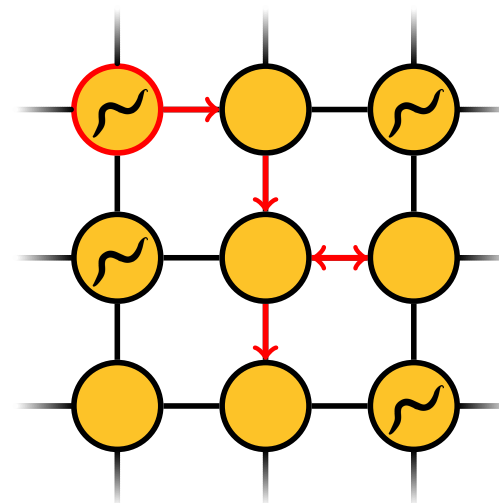
Development



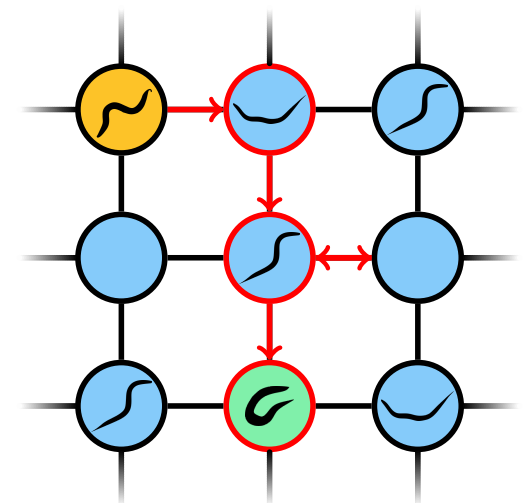
Egg laying



Movement



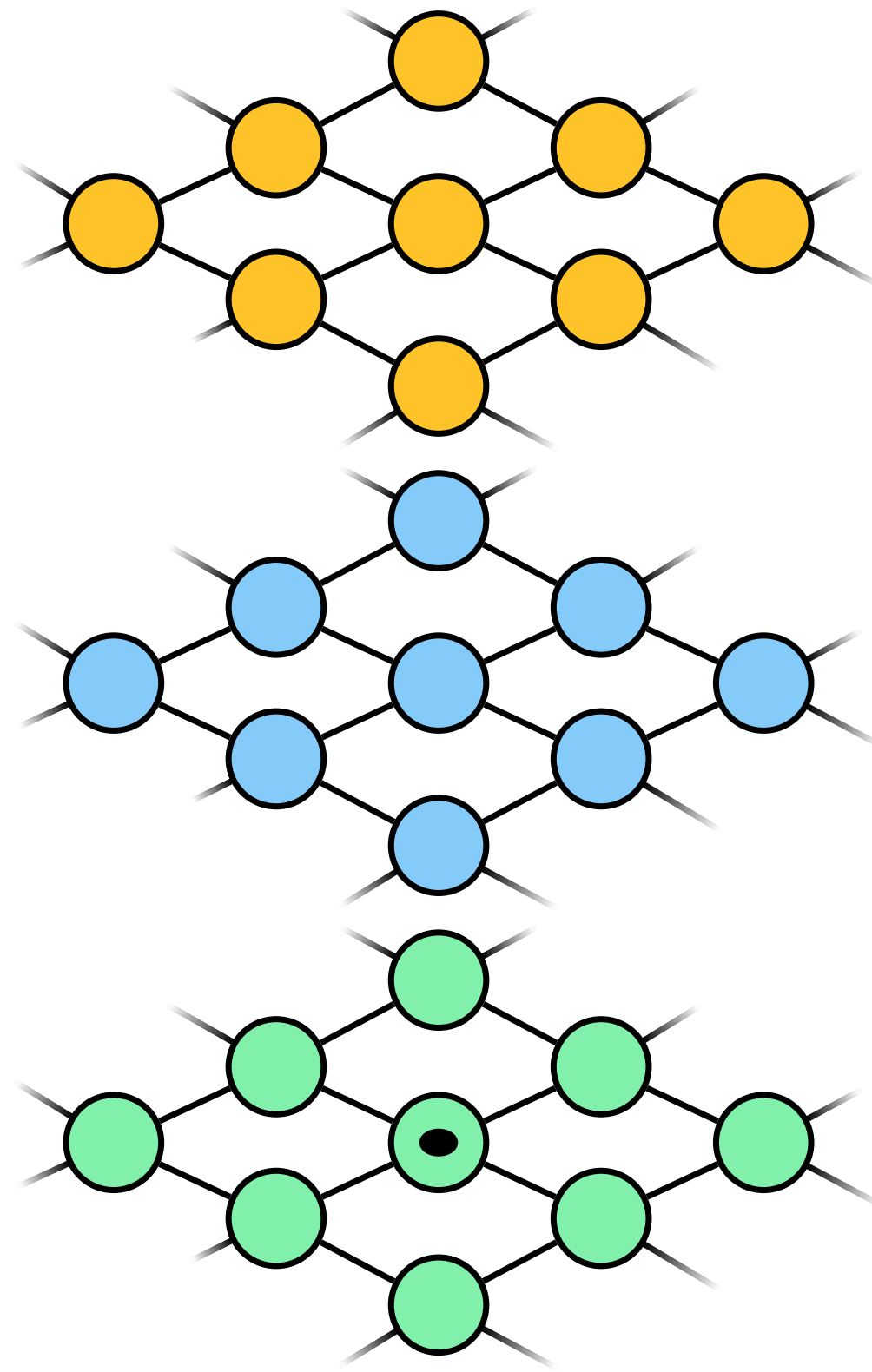
Predation



PriPOP model

Alternative developmental paths

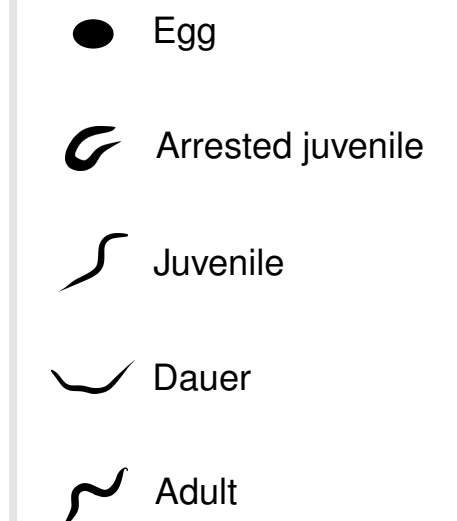
Egg to Juvenile (I)



- Egg
- ⤿ Arrested juvenile
- ⤿ Juvenile
- ⤿ Dauer
- ⤿ Adult

Alternative developmental paths

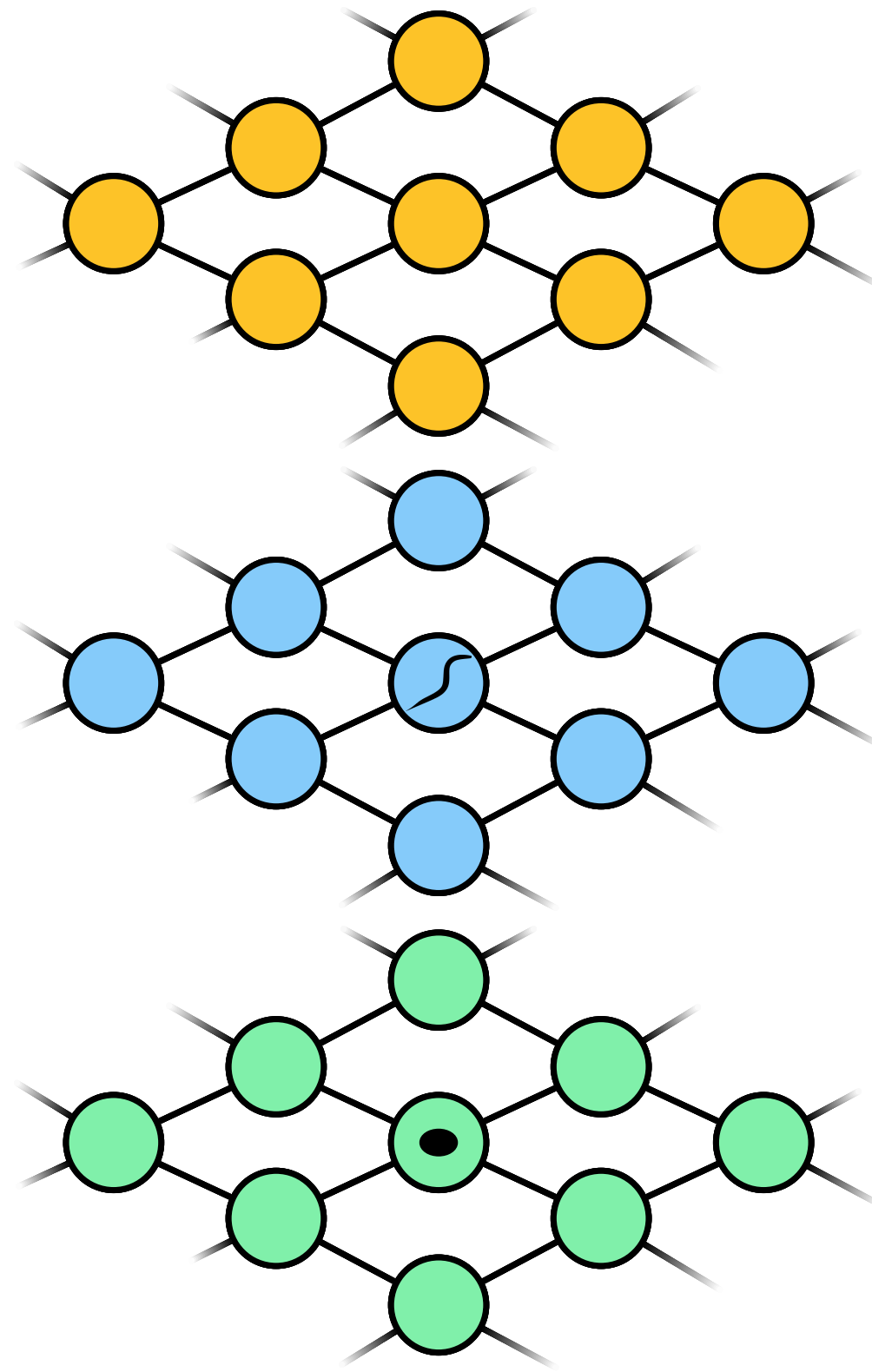
The diagram illustrates a three-layer lattice structure, likely representing a quantum material or a model for topological states. The layers are color-coded: the top layer consists of yellow nodes, the middle layer consists of blue nodes, and the bottom layer consists of green nodes. Each layer is a triangular lattice. A red arrow points from a black dot in the bottom layer to a blue node in the middle layer. The blue node has a wavy line, possibly representing a defect or a specific state. The lattice is connected by black lines, and each node has external lines extending from it.



PriPOP model

Alternative developmental paths

Egg to Juvenile (II)

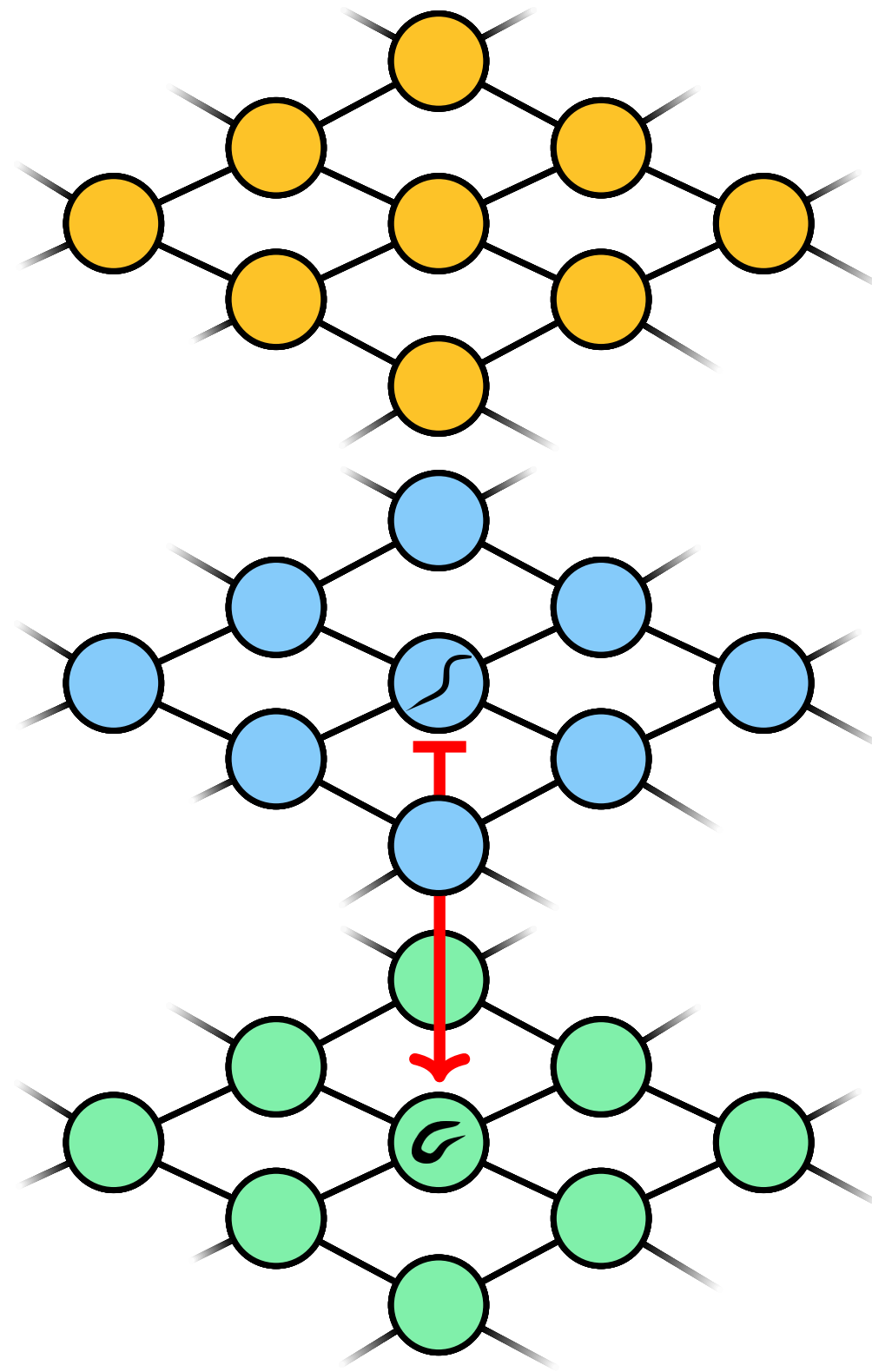


- Egg
- ⤿ Arrested juvenile
- ⤿ Juvenile
- ⤿ Dauer
- ⤿ Adult

PriPOP model

Alternative developmental paths

Egg to Juvenile (II)

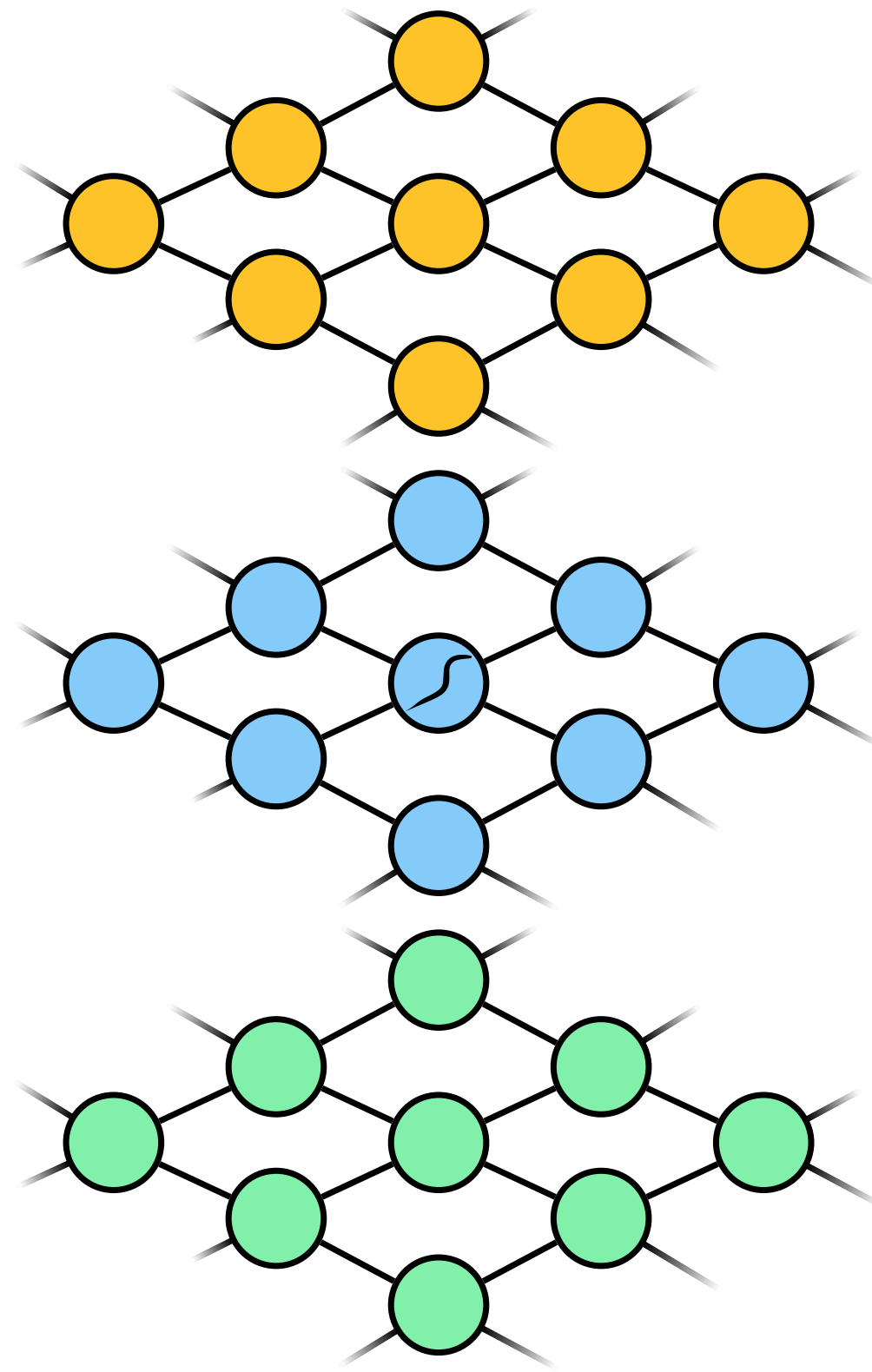


- Egg
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- ⤿ Adult

PriPOP model

Alternative developmental paths

Juvenile to adult (I)

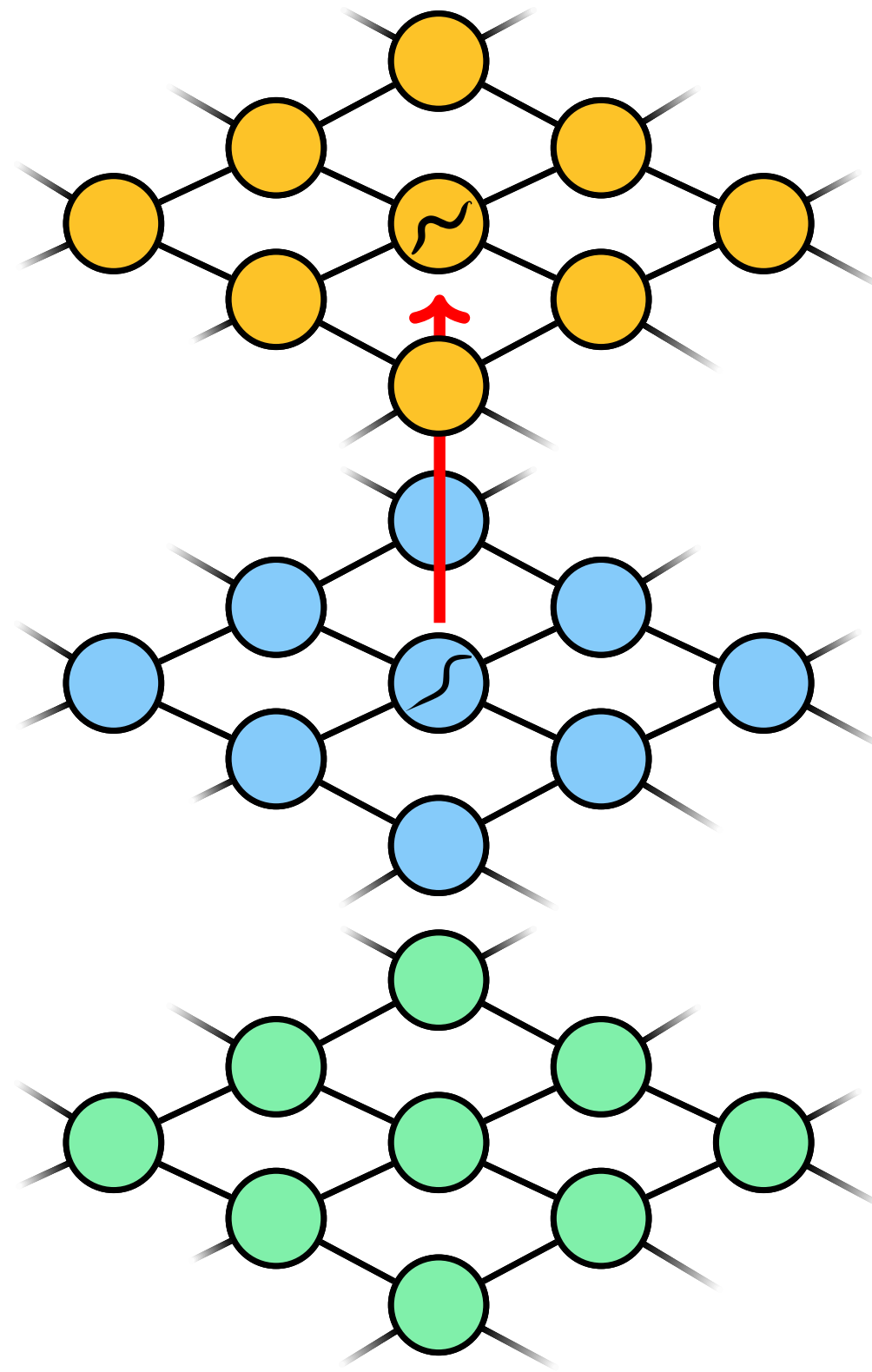


- Egg
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- ⤿ Dauer
- ⤿ Adult

PriPOP model

Alternative developmental paths

Juvenile to adult (I)

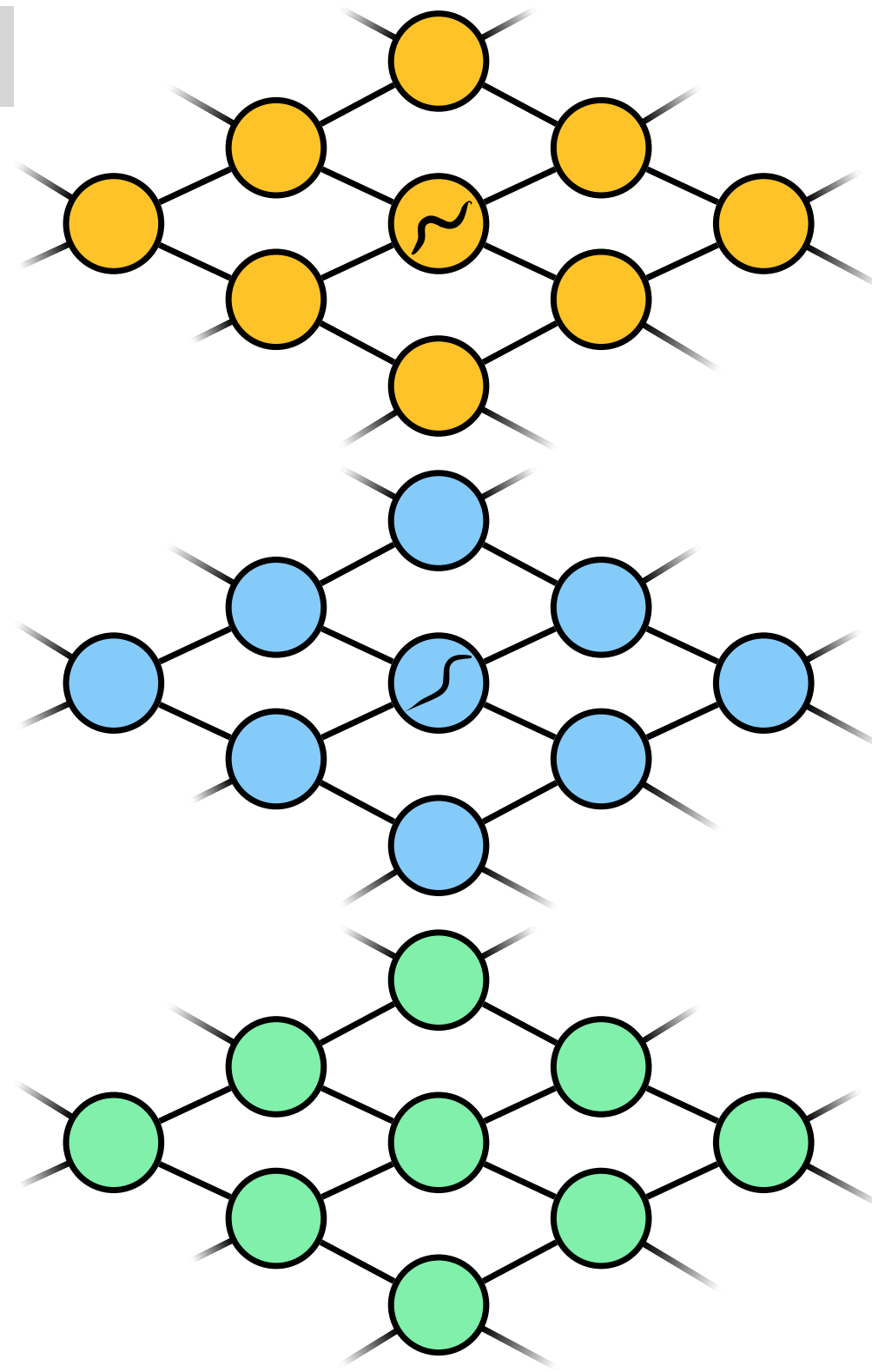


- Egg
- ⤿ Arrested juvenile
- ⤿ Juvenile
- ⤿ Dauer
- ⤿ Adult

PriPOP model

Alternative developmental paths

Juvenile to adult (II)

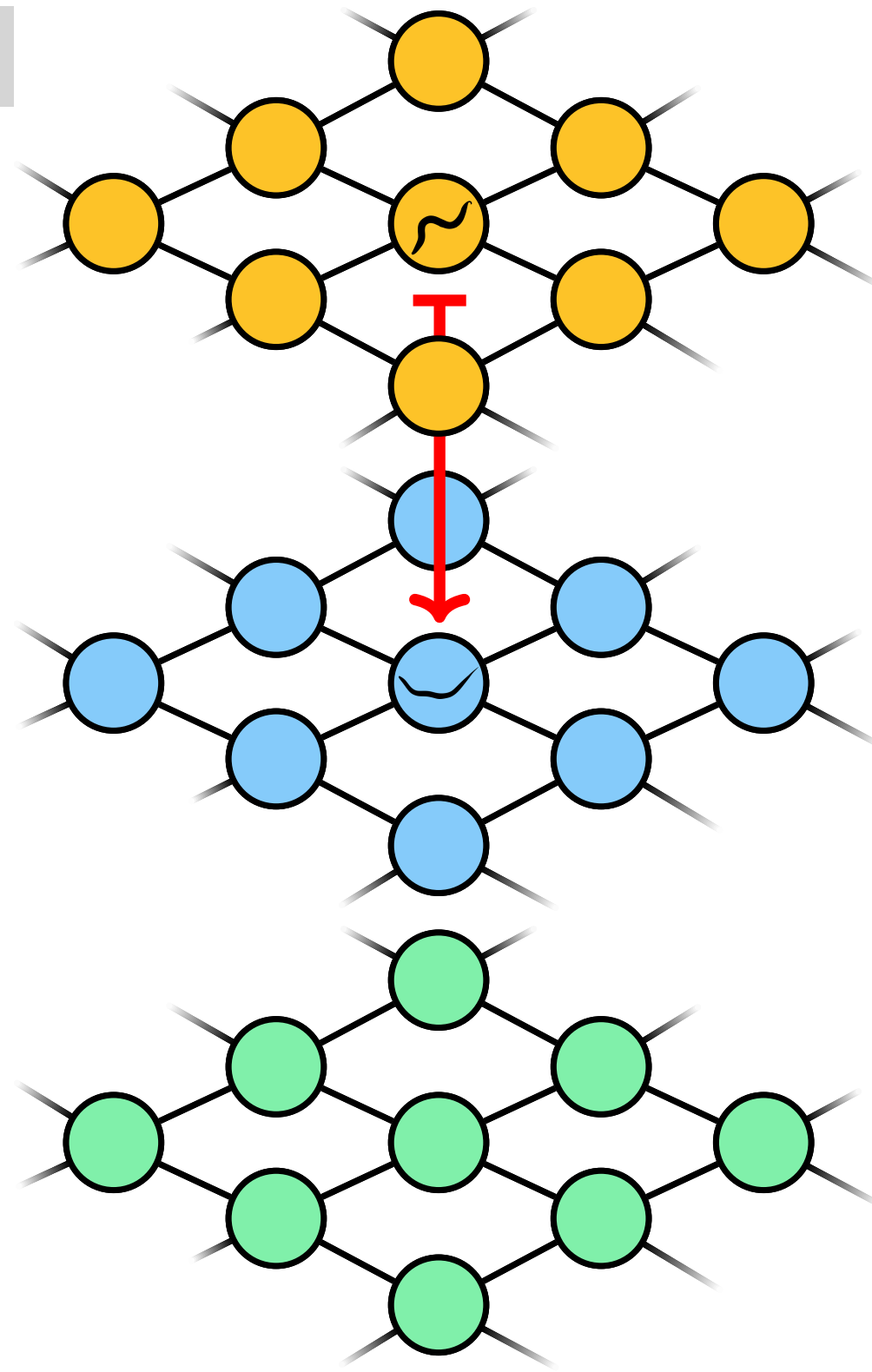


- Egg
- ⤿ Arrested juvenile
- ⤿ Juvenile
- ⤿ Dauer
- ⤿ Adult

PriPOP model

Alternative developmental paths

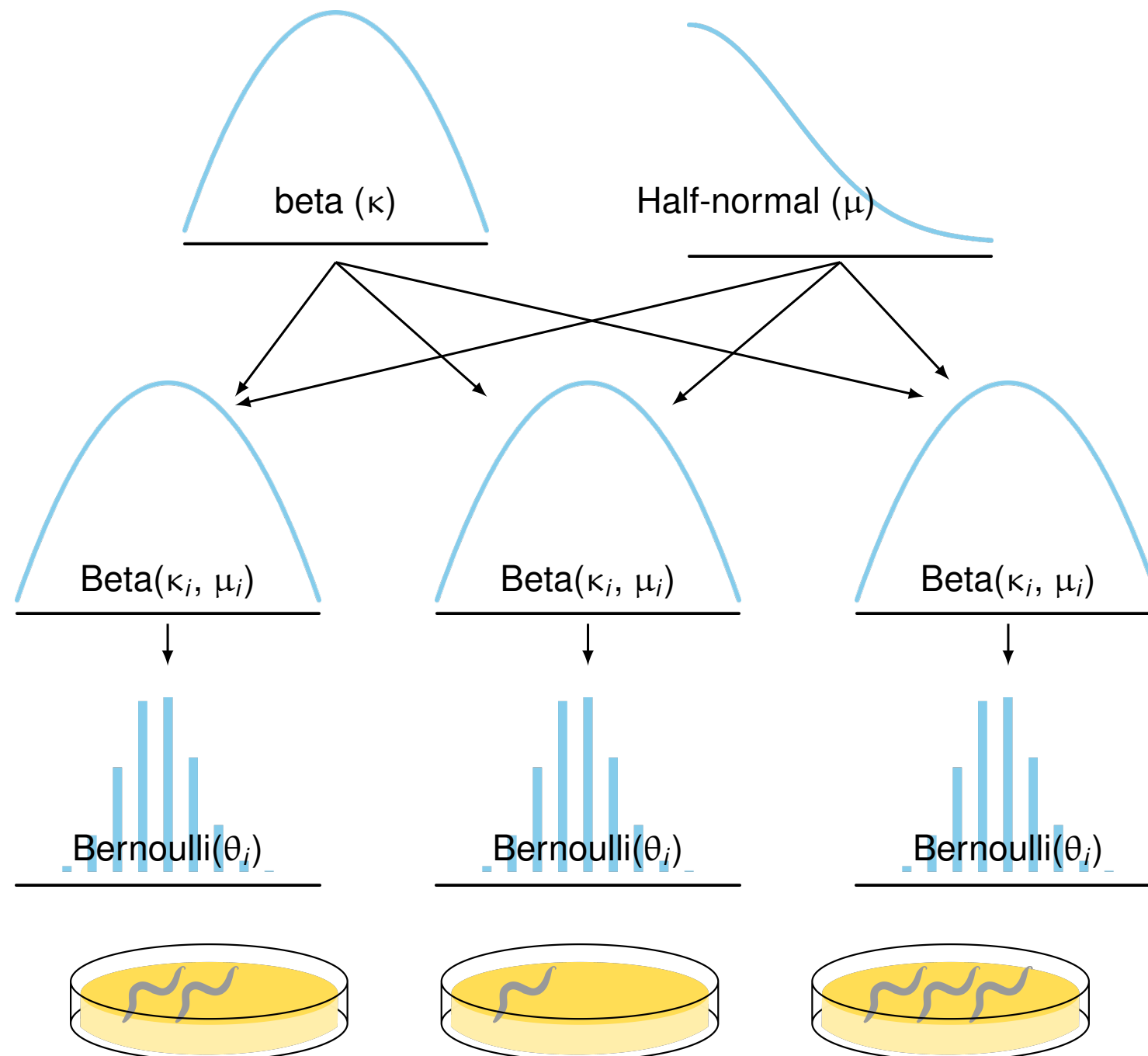
Juvenile to adult (II)



- Egg
- ⤿ Arrested juvenile
- ⤿ Juvenile
- ⤿ Dauer
- ⤿ Adult

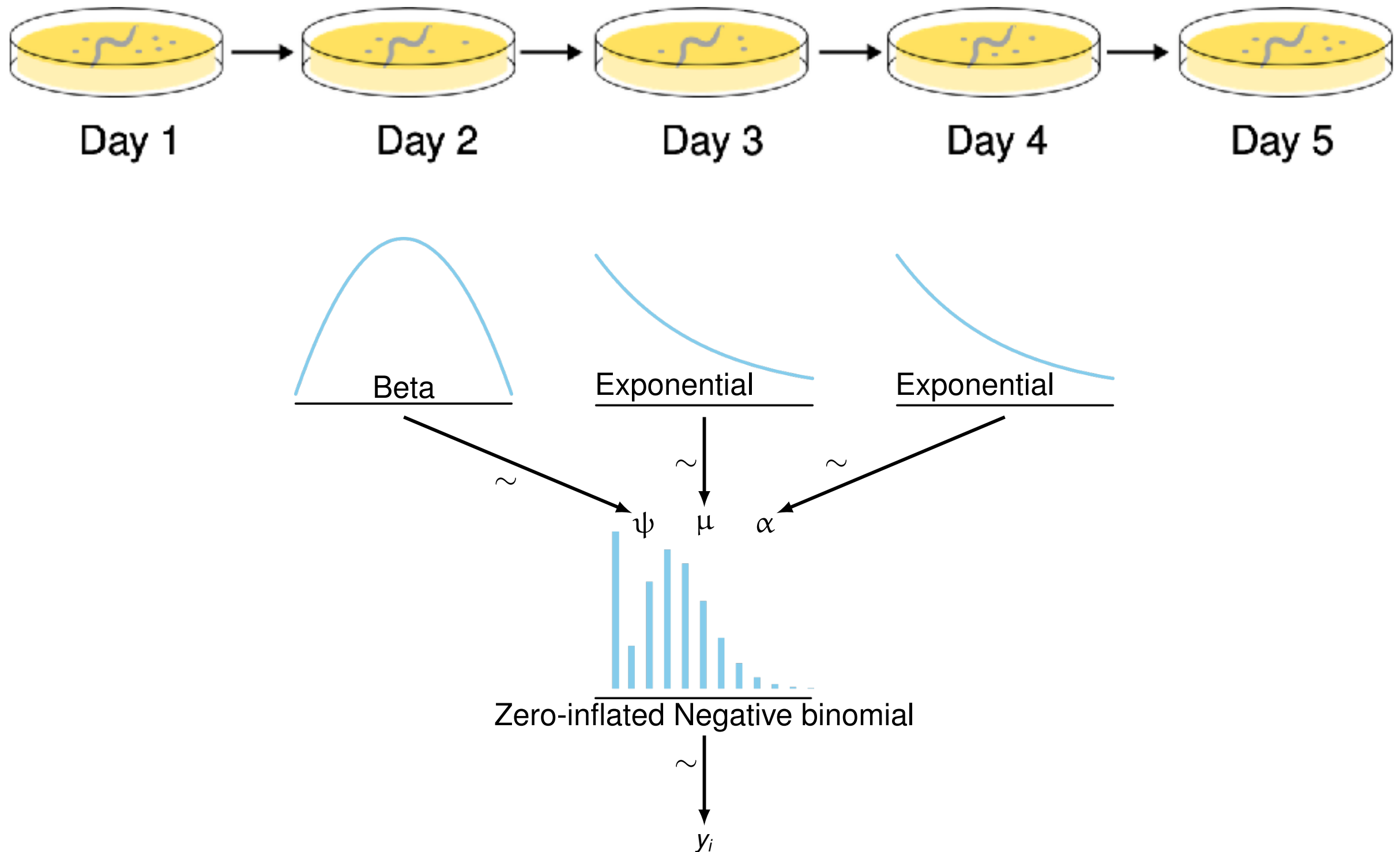
PriPOP model

Capturing the variation in mouth-form plasticity



PriPOP model

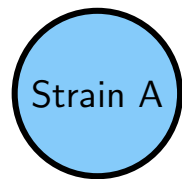
Capturing the variation in fecundity



PriPOP model

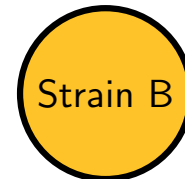
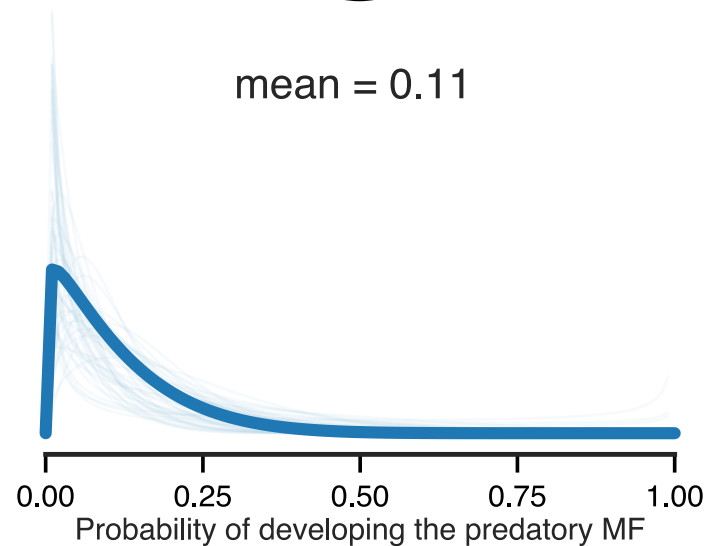
Diet-induced plasticity

E. Coli OP50



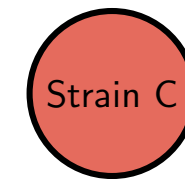
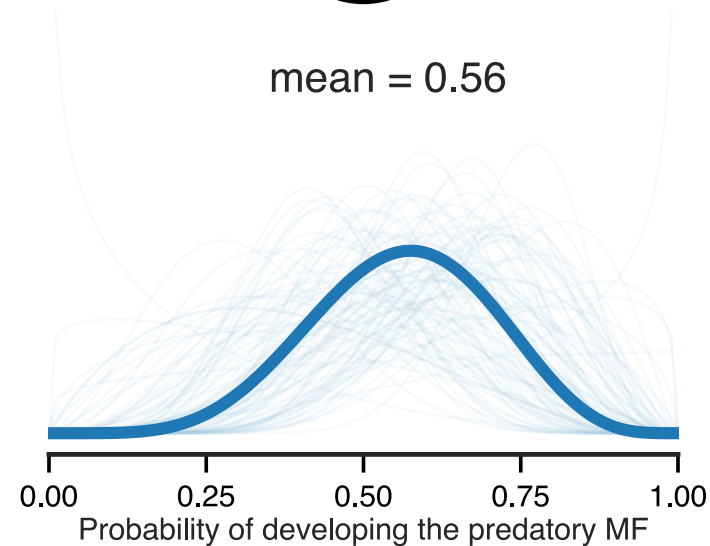
Strain A

mean = 0.11

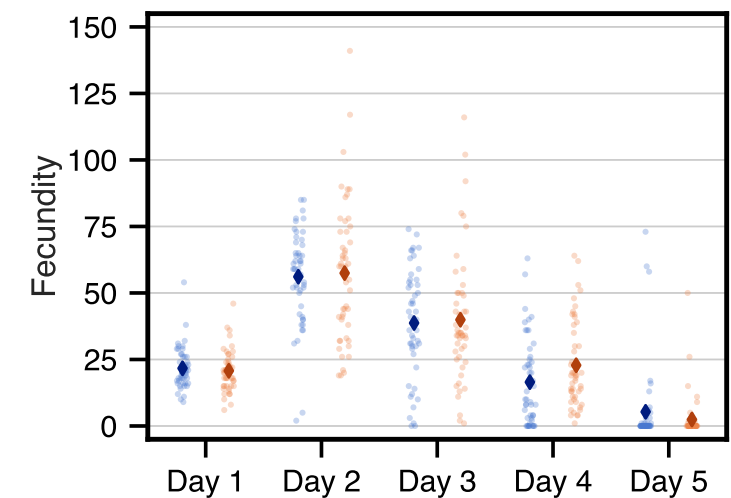
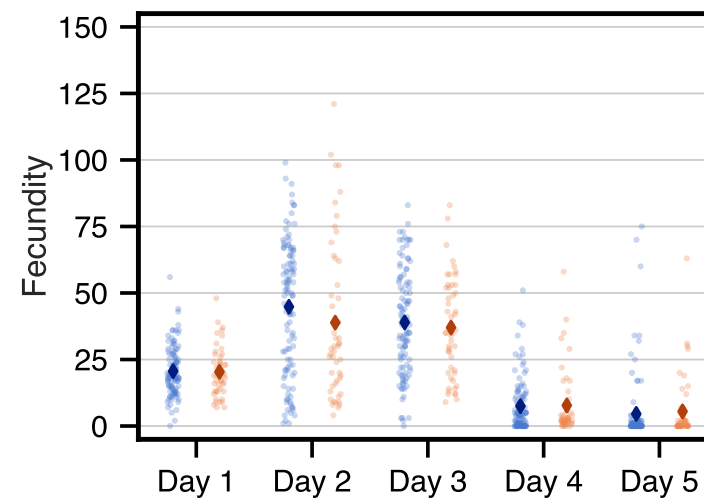
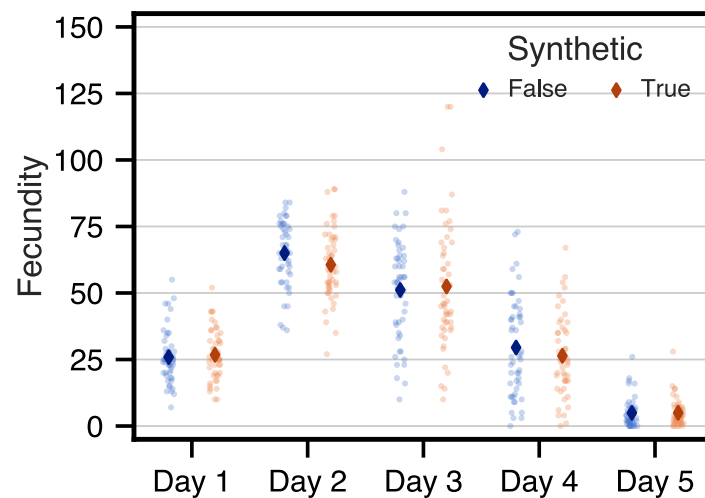
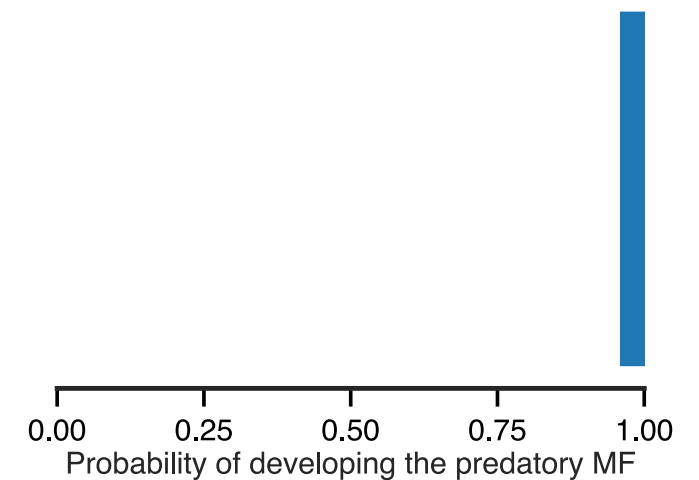


Strain B

mean = 0.56



Strain C

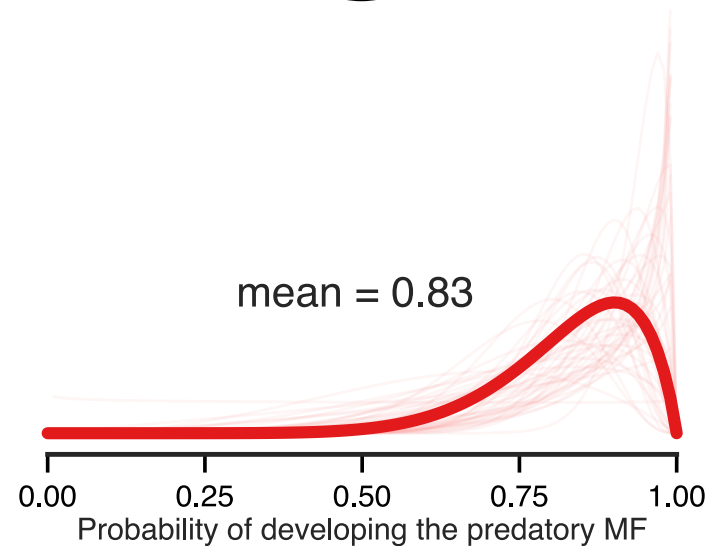


PriPOP model

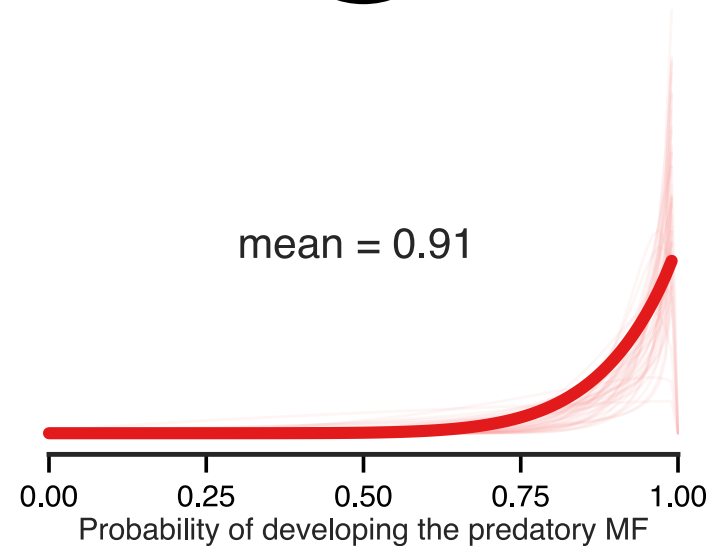
Diet-induced plasticity

Novosphingobium sp. L76

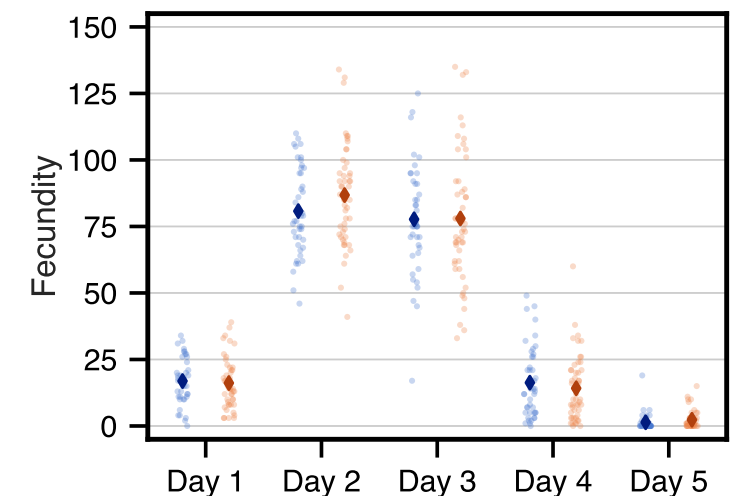
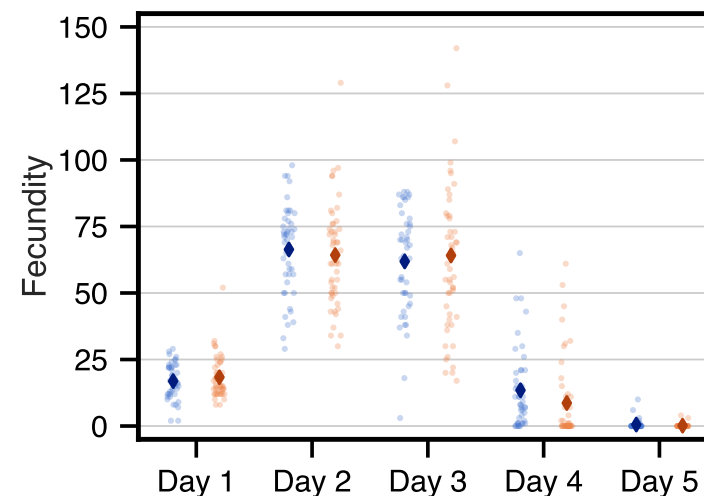
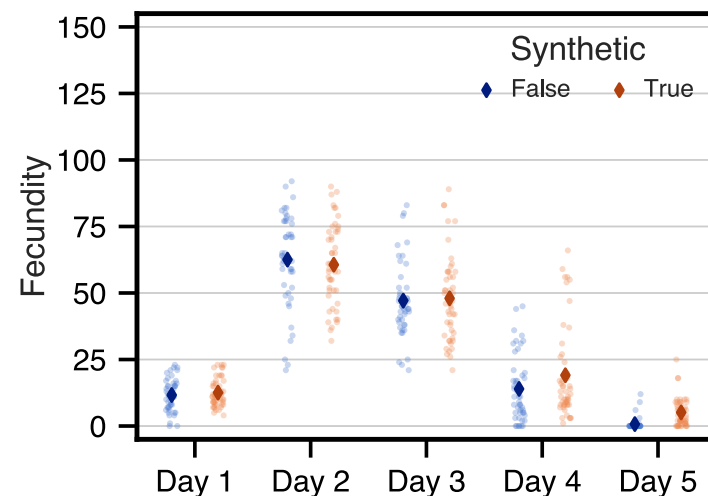
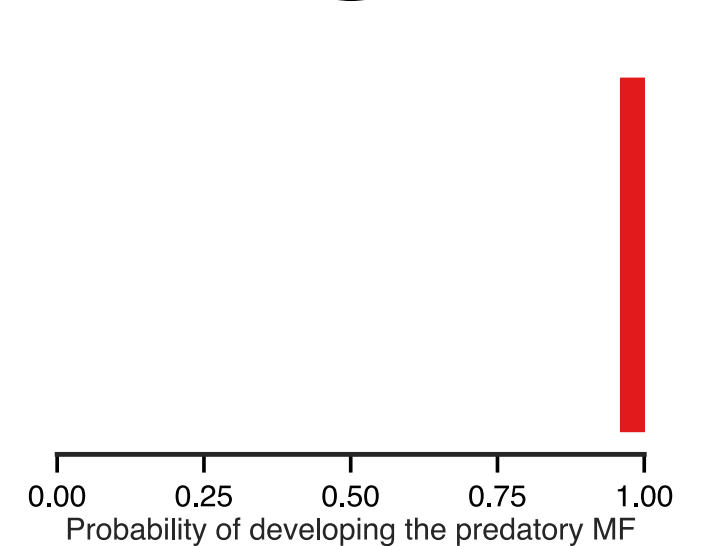
Strain A



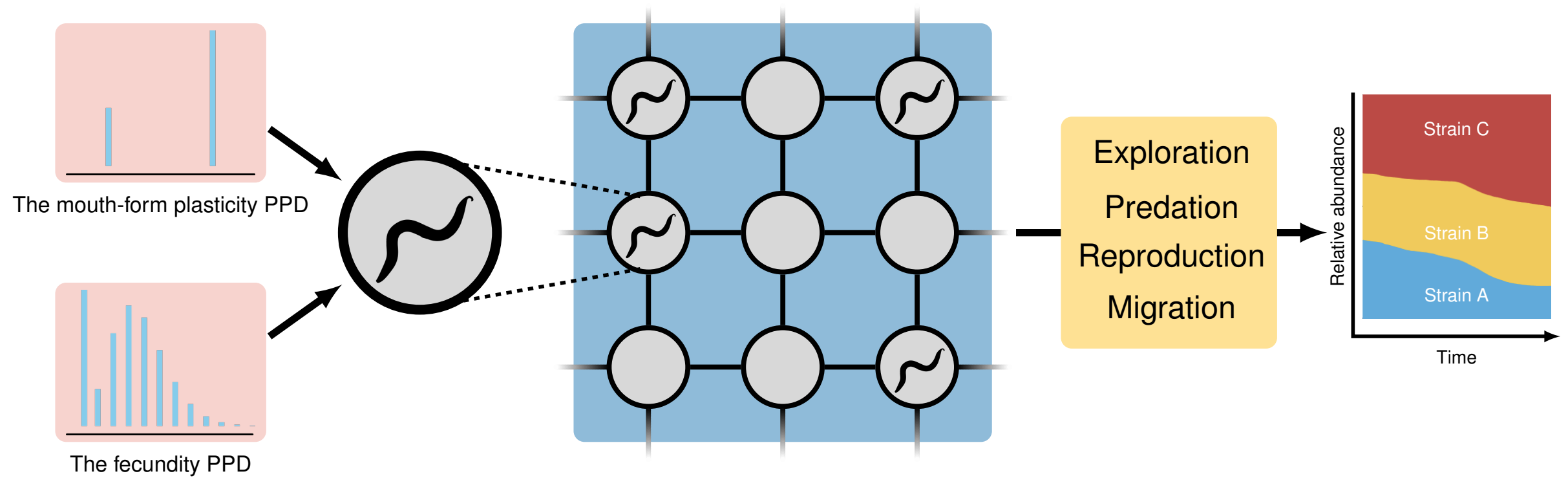
Strain B



Strain C

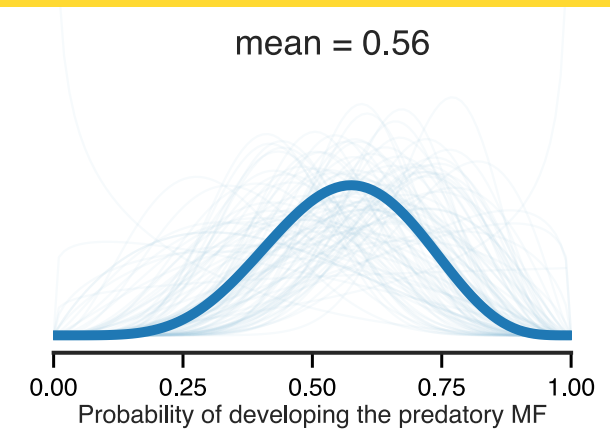
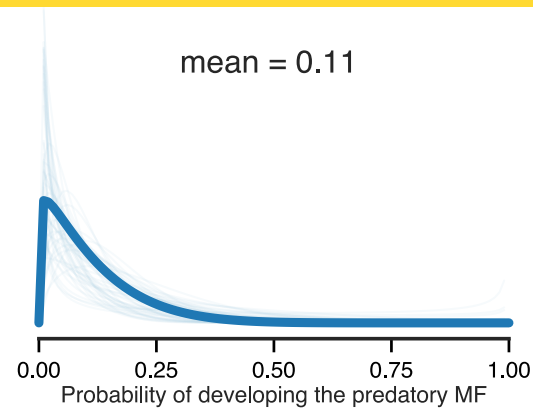
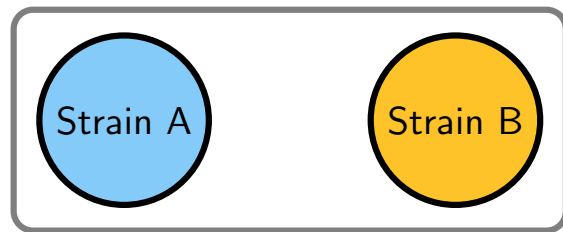


Workflow of *PriPOP*



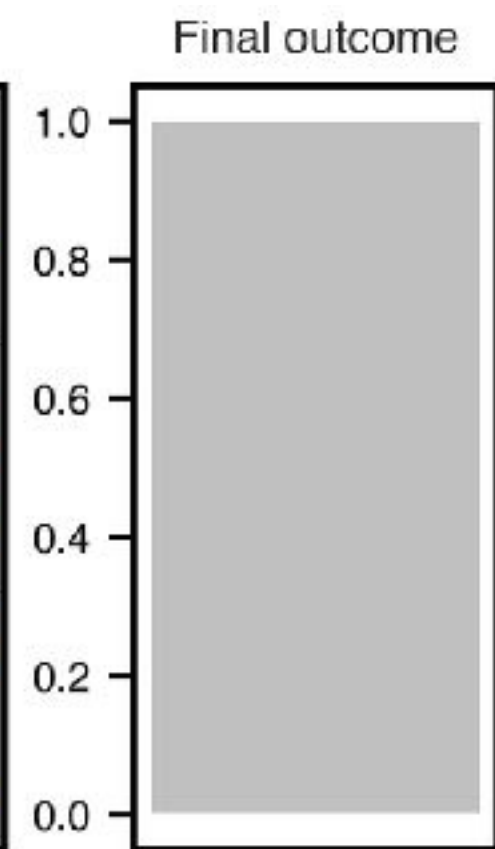
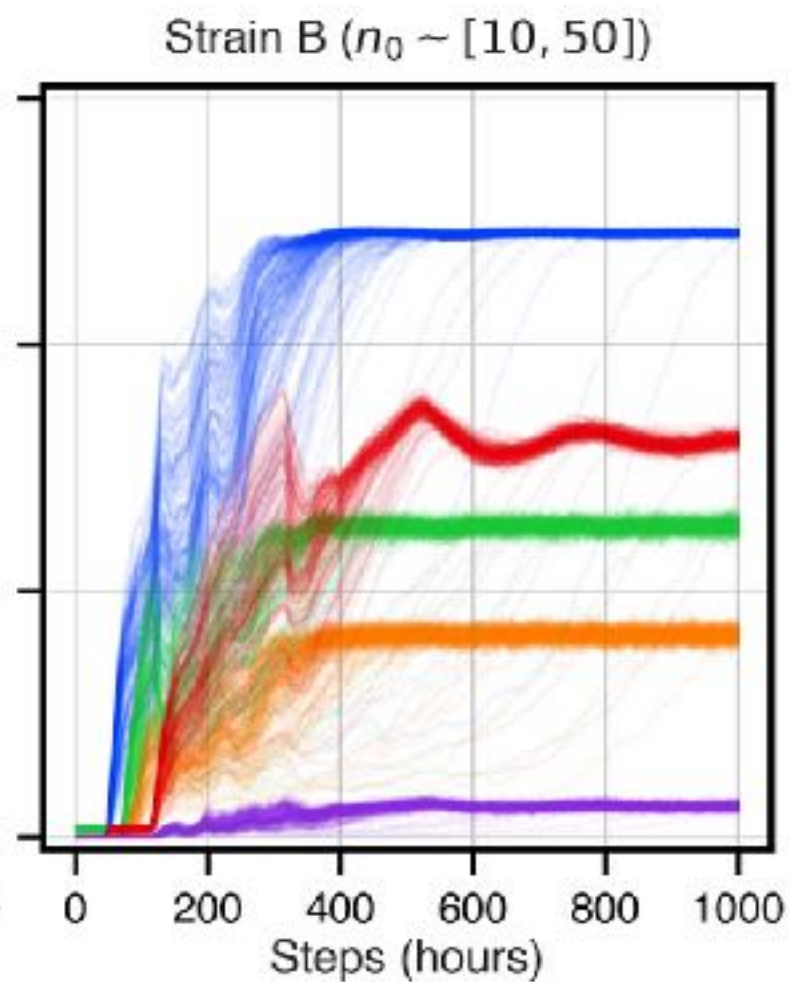
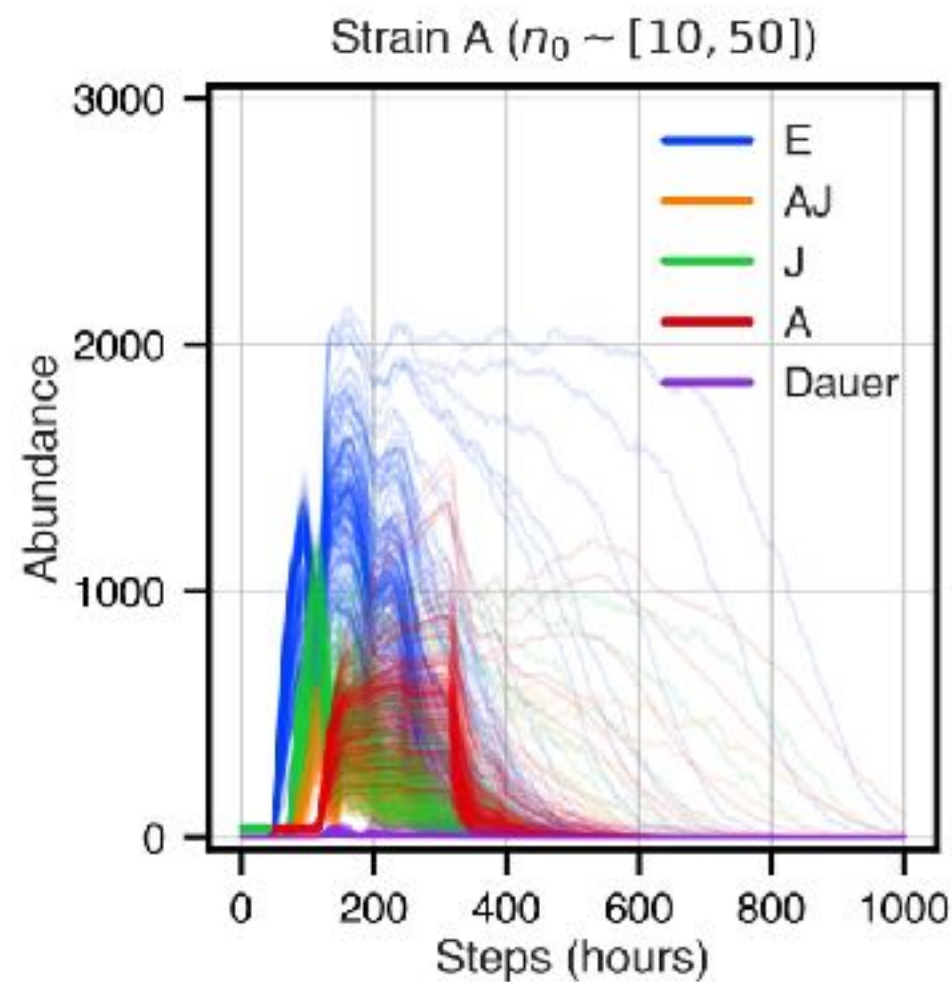
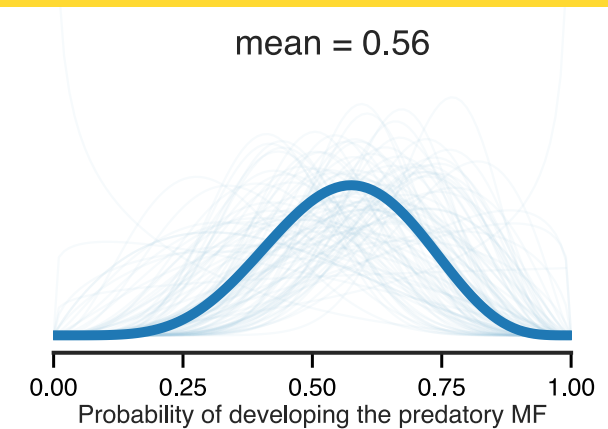
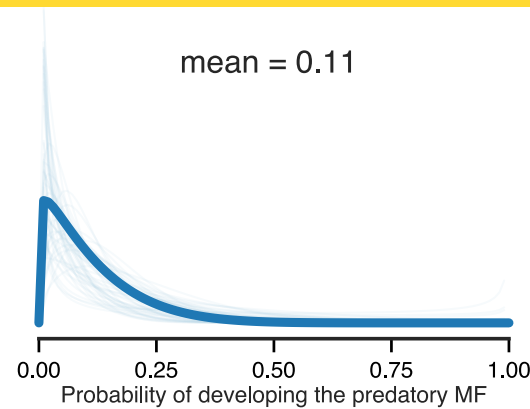
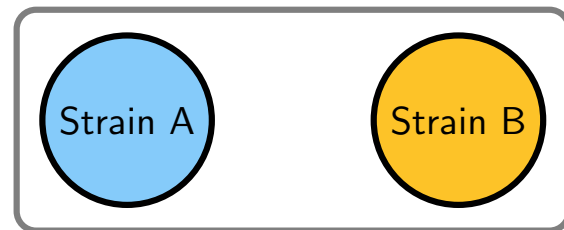
Pairwise competition

E. Coli OP50



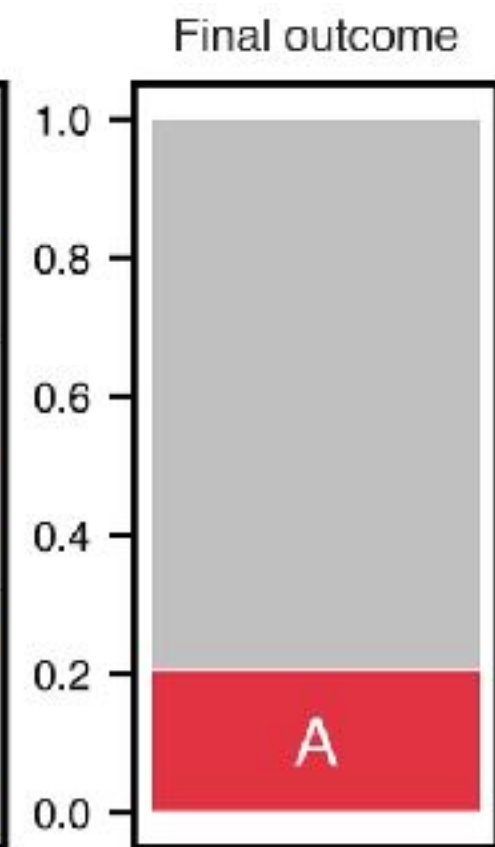
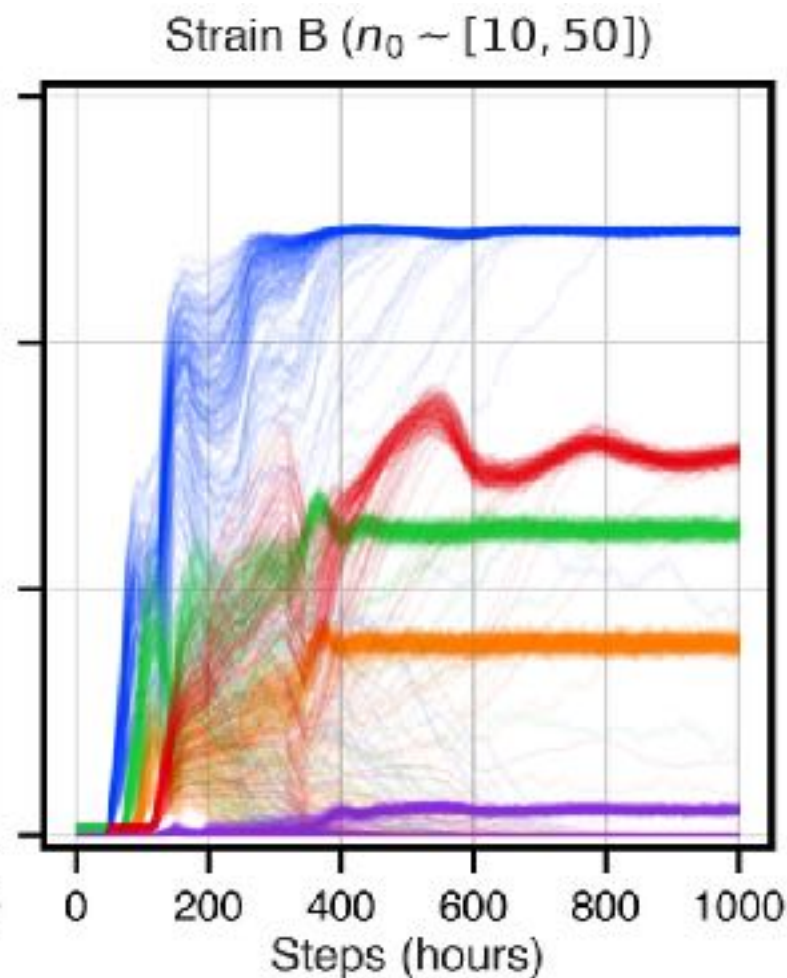
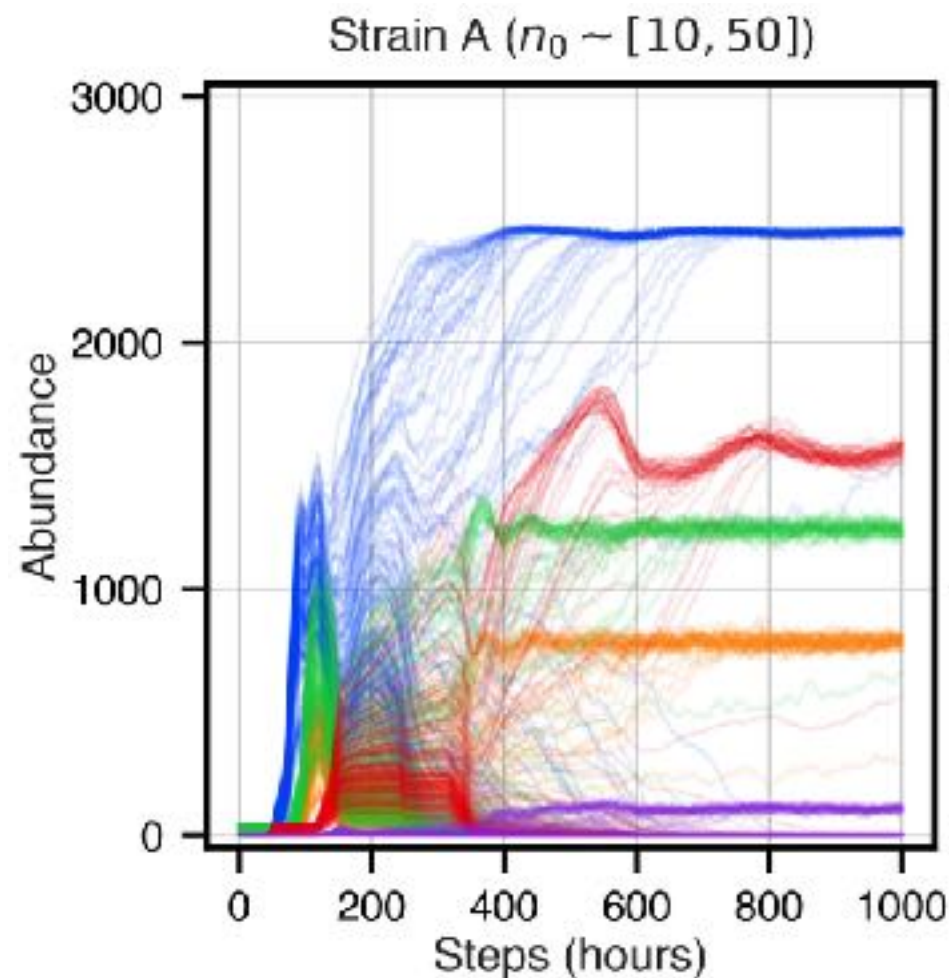
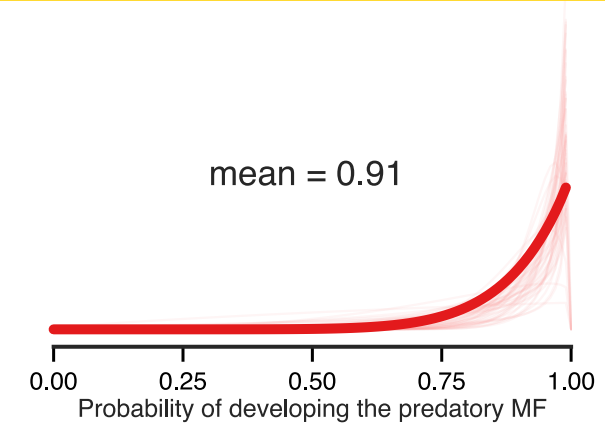
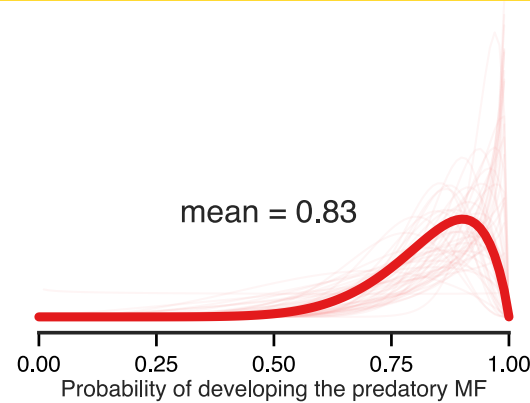
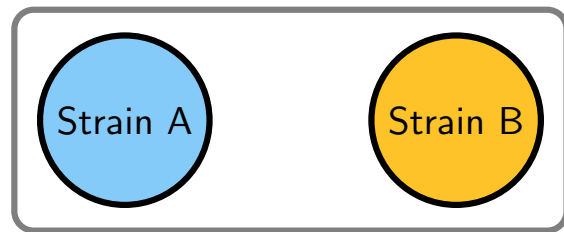
Pairwise competition

E. Coli OP50

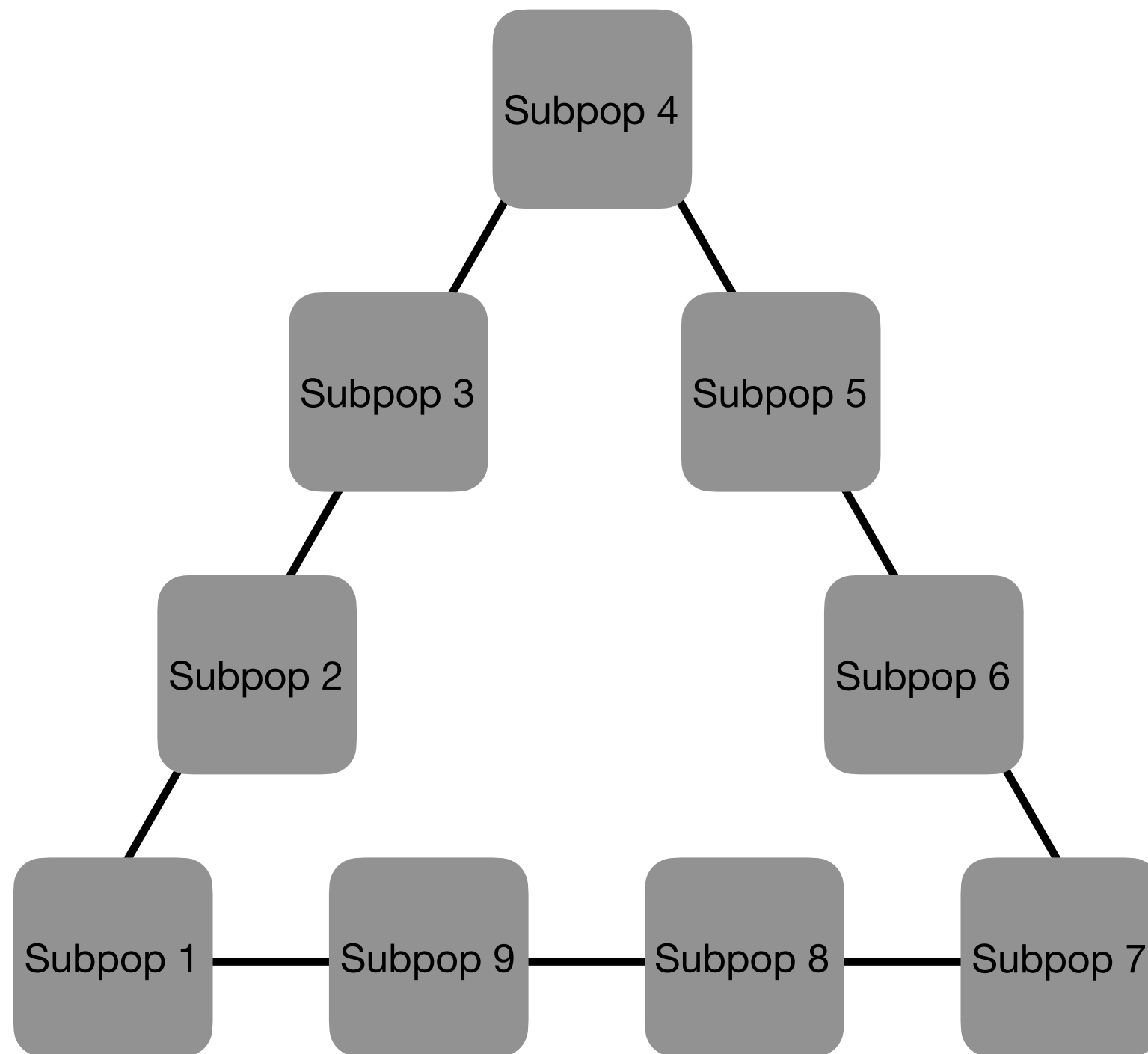


Pairwise competition

Novosphingobium sp. L76

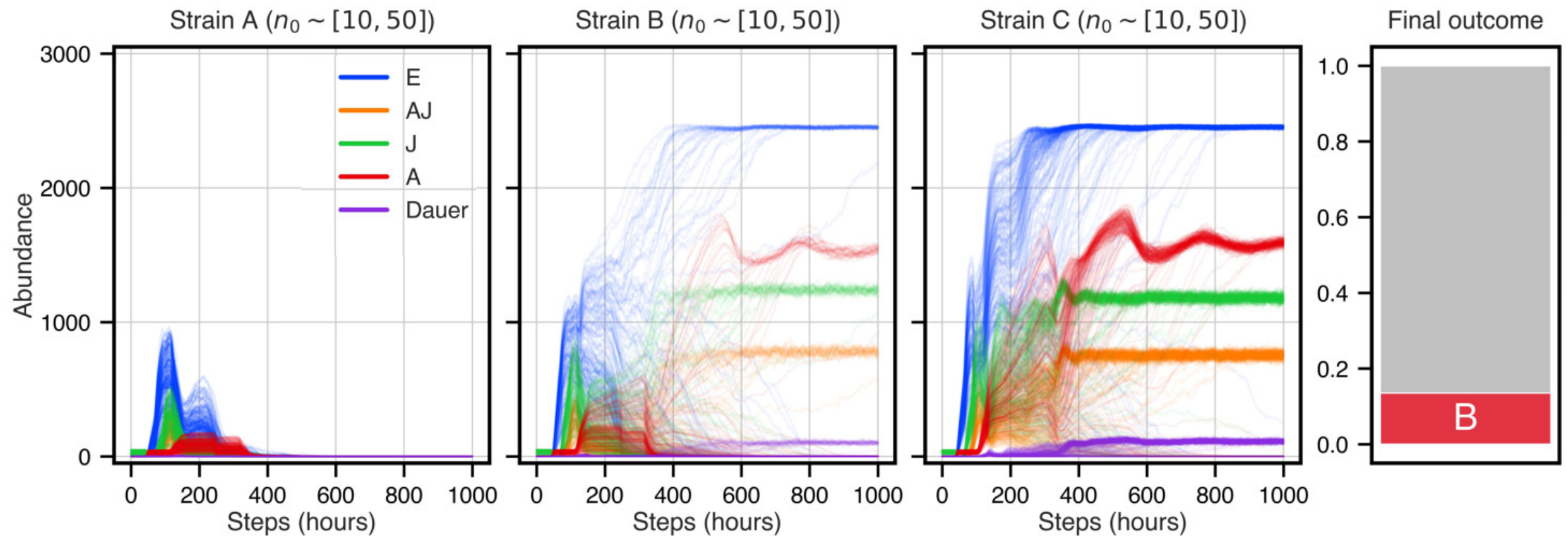


Coexistence in a metapopulation



Coexistence in a population

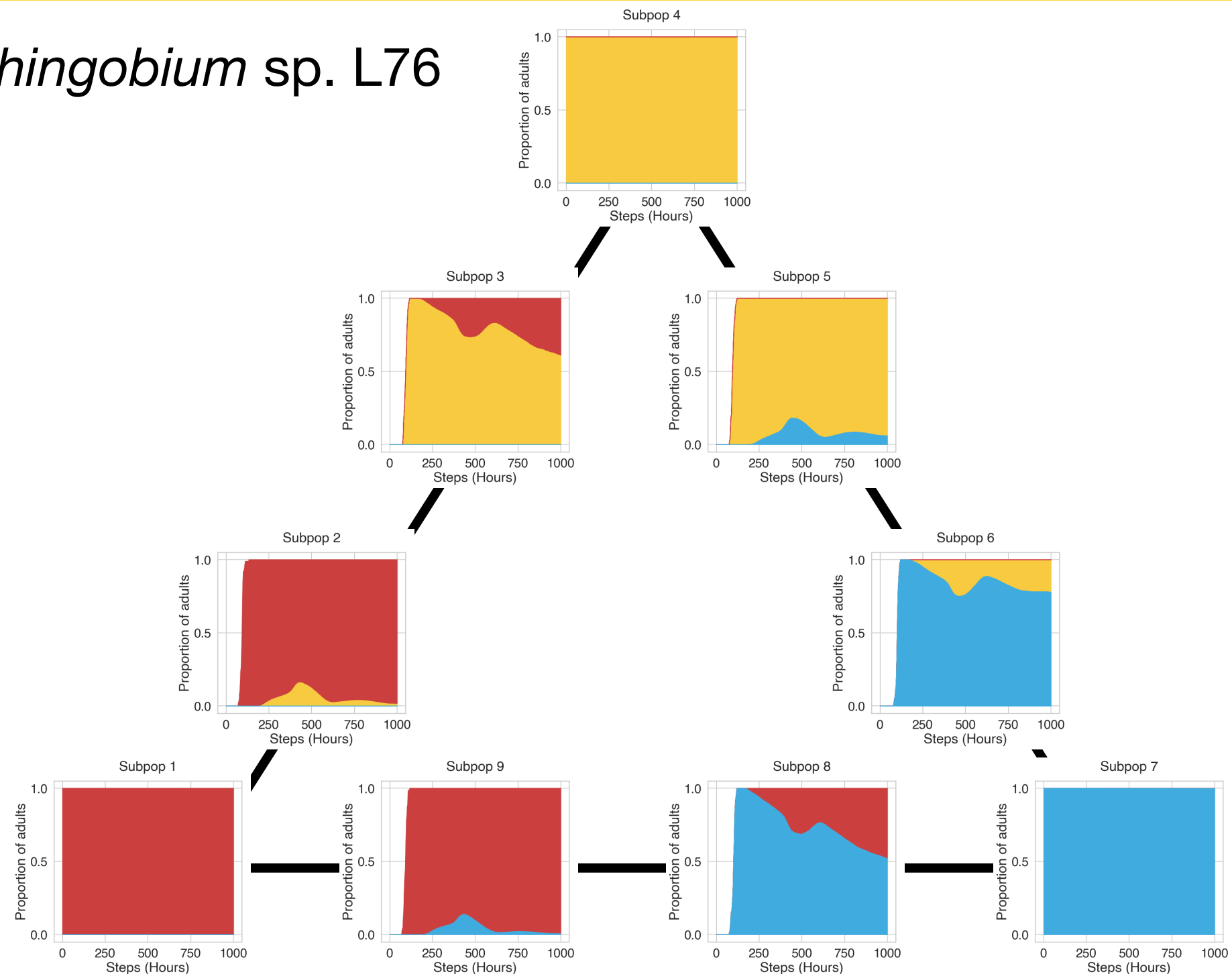
Three-way competition



Coexistence in a metapopulation

Three-way competition

Novosphingobium sp. L76



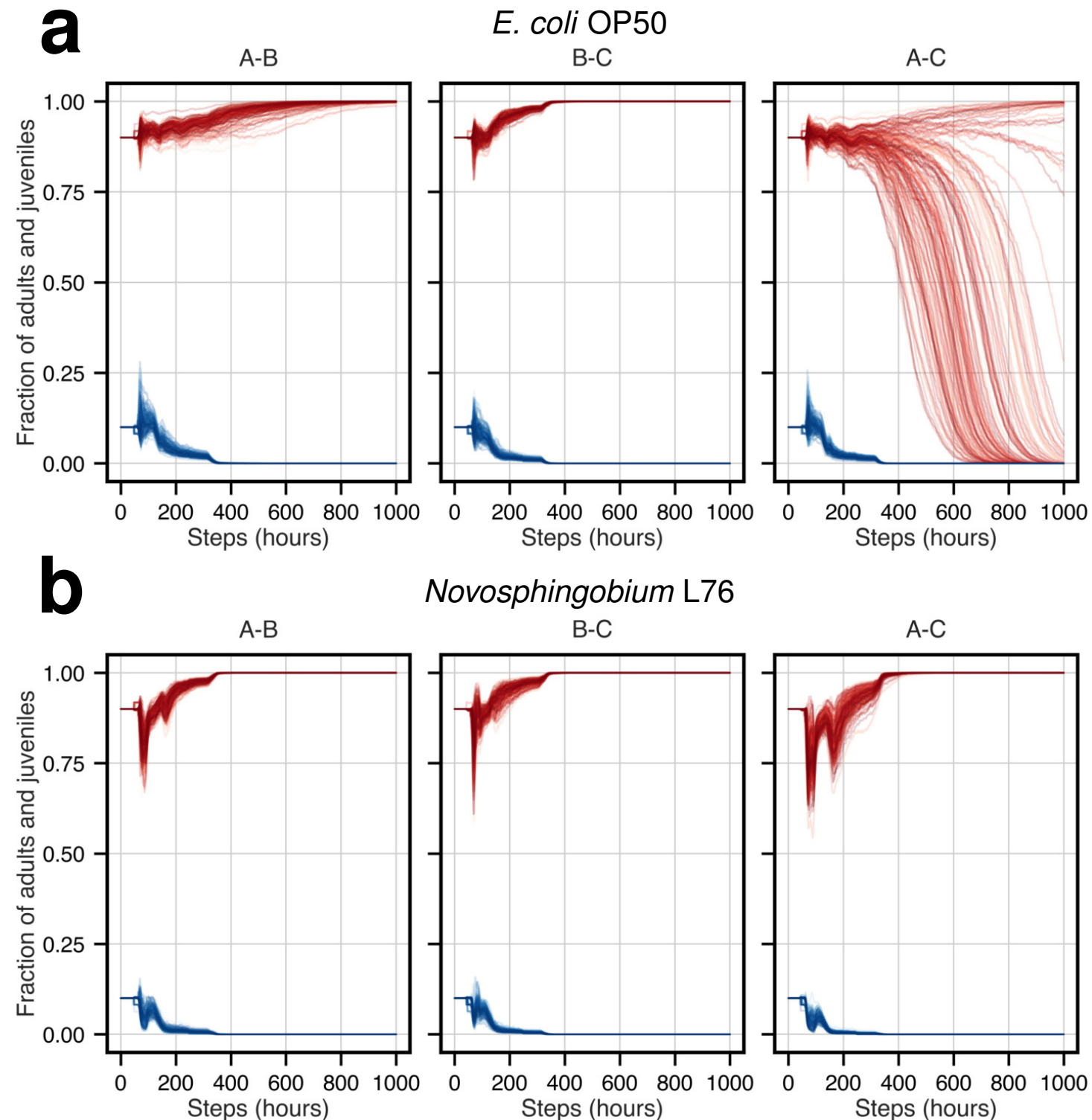
Strain A

Strain B

Strain C

Coexistence in a metapopulation

Invasibility



The theatre of ecology

Individual-level measurements

Phenotypic plasticity
Non-genetic trait variation
Trade-offs between traits



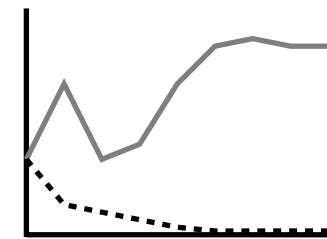
Ecological *mise-en-scène*

Biotic interactions
Spatial heterogeneity
Temporal heterogeneity

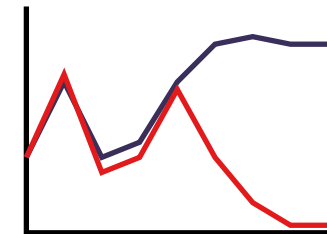


Ecological outcomes

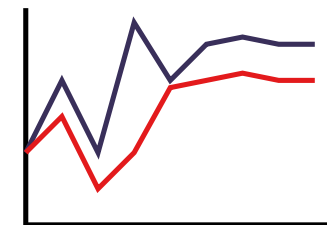
Competitive exclusion



Transient coexistence



Long-term coexistence



Stochasticity, plasticity and coexistence

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The role of plasticity and stochasticity in coexistence

Ata Kallrad, Ralf J. Sommer 

First published: 13 February 2024 | <https://doi.org/10.1111/ele.14370>

Editor: Bernd Blasius

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Abstract

Species coexistence in ecological communities is a central feature of biodiversity. Different concepts, i.e., contemporary niche theory, modern coexistence theory, and the unified neutral theory, have identified many building blocks of such ecological assemblies. However, other factors, such as phenotypic plasticity and stochastic inter-individual variation, have received little attention, in particular in animals. For example, how resource polyphenisms resulting in predator–prey interactions affect coexistence is currently unknown. Here, we present an integrative theoretical–experimental framework using the nematode plasticity model *Pristionchus pacificus* with its well-studied mouth-form dimorphism resulting in cannibalism. We develop an individual-based model that relies upon synthetic data based on our empirical measurements of fecundity and polyphenism to preserve demographic heterogeneity. We demonstrate how the interplay between plasticity and individual stochasticity result in all-or-nothing outcomes at the local level. Coexistence is made possible when spatial structure is introduced.

Acknowledgments



Ralf Sommer
MPI for Biology Tübingen



Steve Pennings
University of Houston



Mohannad Dardiry
Ikmi group at EMBL



Veysi Piskobulu
Sommer Lab

MAX PLANCK
GESELLSCHAFT



 UNIVERSITY OF HOUSTON

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